NBA Data EDA

Cross-Season Analysis for Predictive Modeling

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Executive Summary

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This exploratory data analysis examines five seasons of NBA player performance data (2021-22 through 2025-26), comprising 136,965 player-game observations across approximately 1,000 unique players and 5,274 games. The analysis investigates the predictability of player performance for applications in sports betting and prediction markets, specifically player props (points, rebounds, assists over/under lines) and team outcomes (moneyline, spread, totals).

The central finding is that a player's season average alone explains approximately 50% of the variance in single-game point totals (R^2 = 0.50). This relationship remains remarkably stable across all five seasons, with R^2 values ranging from 0.49 to 0.54. Star players (those averaging 20+ points per game) exhibit substantially more consistent performance than bench players, with coefficients of variation of 0.36 versus 0.95 respectively.

These patterns suggest that player-level features will dominate any predictive model, while contextual features such as home court advantage (55.1% win rate, Cohen's h = 0.10) and rest days contribute only marginally. The stability of these relationships across seasons supports the development of robust predictive models, though the 2025-26 partial season data should be used cautiously given it represents only 677 games through February 2026.

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Chapter 1

Introduction

The growing sophistication of sports betting markets and the emergence of prediction platforms such as Polymarket create opportunities for quantitative approaches to sports prediction. This report presents an exploratory data analysis of NBA player performance data, examining patterns that may inform predictive modeling for player proposition bets and team outcomes.

Player proposition bets, commonly called "player props," allow bettors to wager on individual statistical outcomes. For example, a sportsbook might set LeBron James's points line at 24.5, meaning bettors can wager on whether he will score over or under that number. Understanding the distributional properties of player performance—its central tendency, variance, and predictability—is fundamental to identifying value in such markets.

1.1 Scope of Analysis

This analysis encompasses five NBA seasons from 2021-22 through the partial 2025-26 season, totaling 136,965 player-game observations. The dataset includes traditional boxscore statistics (points, rebounds, assists, field goal percentages), advanced metrics (true shooting percentage, usage rate, offensive and defensive ratings), and pregame contextual information (rolling averages, team statistics, rest days).

The primary prediction targets investigated are:

- **Player Props:** Points (PTS), rebounds (REB), and assists (AST) per game
- **Team Outcomes:** Win/loss (moneyline), point differential (spread), and combined score (totals)

1.2 Key Questions

The analysis addresses three fundamental questions relevant to prediction market applications:

1. What makes player performance predictable? Which features correlate most strongly with single-game outcomes, and how does predictability vary across player types?
2. Do predictive patterns hold across seasons? Are the relationships observed in historical data stable enough to support forward-looking models?

3. How has the NBA evolved? Have changes in playing style, scoring pace, or other league-wide trends affected the distributional properties relevant to prediction?

Betting Terminology

Throughout this report, several sports betting terms appear:

- **Player Props (Proposition Bets):** Bets on individual player statistical outcomes rather than game results. For example, "LeBron James Over 24.5 Points" is a player prop—you win if LeBron scores 25 or more, lose if he scores 24 or fewer. Player props can target any statistic: points, rebounds, assists, three-pointers made, steals, blocks, or combinations thereof. They represent our primary prediction target in this analysis.
- **Over/Under (O/U):** A bet on whether a statistic will exceed or fall below a specified line
- **Moneyline:** A bet on which team will win the game outright
- **Spread:** A bet on the margin of victory, with points given to the underdog
- **Totals:** A bet on the combined score of both teams

1.3 Report Structure

Chapter 2 describes the dataset, including its dimensions, quality, and the distribution of key variables. Chapter 3 explains the statistical methods employed throughout the analysis. Chapter 4 presents the core analytical findings, organized by topic and including cross-season comparisons for each finding. Chapter 5 synthesizes the key patterns and their implications for modeling. Chapter 6 acknowledges the limitations of this analysis, and Chapter 7 provides recommendations for subsequent predictive modeling work.

Chapter 2

Data Overview

2.1 Dataset Description

The analysis draws from the `combined_player_stats` table hosted on Railway PostgreSQL, which aggregates boxscore data with pregame features computed from historical performance. Each row represents a single player's statistics for a single game, creating a panel dataset with repeated observations for each player across games and seasons.

Table 2.1: Dataset dimensions by season. The 2025-26 season is partial, covering games through February 2026.

Season	Observations	Games	Players	Date Range
2021-22	29,103	1,144	618	Oct 2021 – Jun 2022
2022-23	29,632	1,156	547	Oct 2022 – Jun 2023
2023-24	28,244	1,073	585	Oct 2023 – Jun 2024
2024-25	32,349	1,224	577	Oct 2024 – Jun 2025
2025-26	17,637	677	541	Oct 2025 – Feb 2026
Total	**136,965**	**5,274**	**$\sim$1,000**	

The feature set comprises approximately 80 columns organized into logical groups. Identifier columns include game ID, player ID, player name, team abbreviation, game date, and season. Boxscore statistics cover the standard categories: points, rebounds, assists, steals, blocks, turnovers, personal fouls, and shooting percentages for field goals, three-pointers, and free throws. Advanced metrics include true shooting percentage, usage rate, offensive and defensive ratings, net rating, and player impact estimate (PIE).

Pregame features, computed prior to each game, fall into two categories. Player-level pregame features include season averages and rolling averages over the last 5 and 10 games for all boxscore statistics. Team-level pregame features include the team's season win percentage, points per game, whether the team is playing at home, and days of rest since the previous game. Finally, quarter-level statistics capture performance by period for games where this granular data is available.

2.2 Data Quality Assessment

Missing data patterns differ systematically across feature groups. Boxscore statistics are essentially complete, with missing values occurring only for players who recorded a "Did Not Play" (DNP) status. Advanced metrics show similar completeness for players who participated in games. Pregame player averages are structurally missing for the first games of new players or those returning from extended absences, as these features require historical data to compute. Pregame team features show minimal missingness after the first few games of each season.

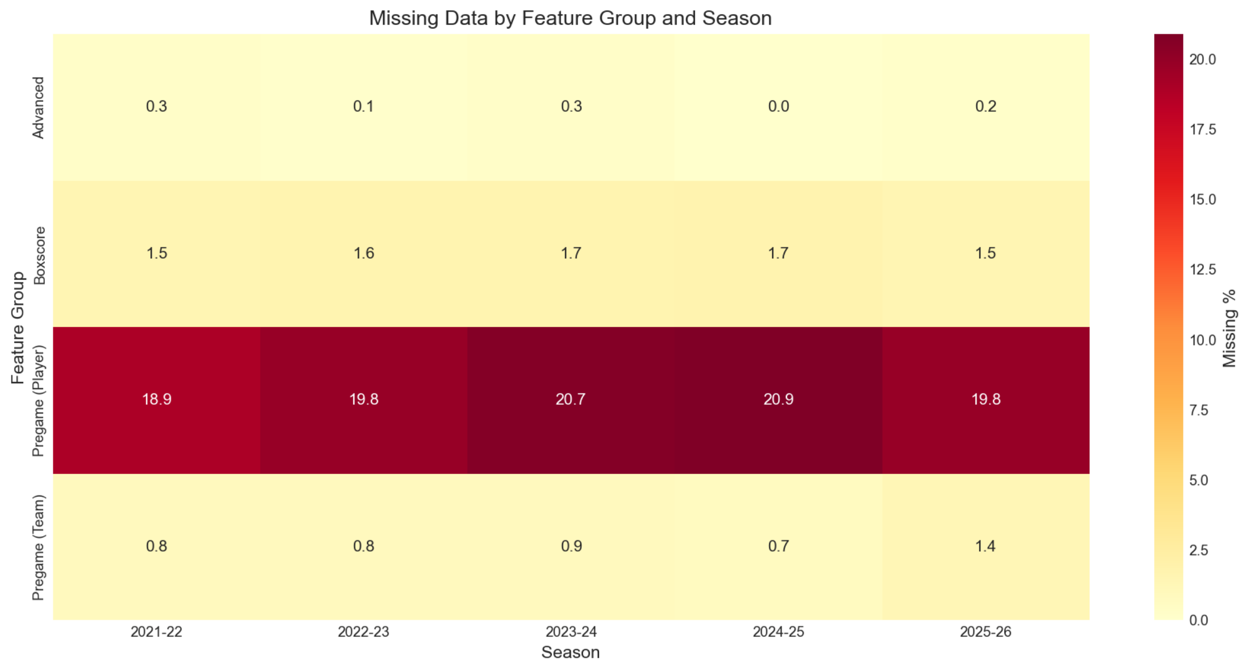

Figure 2.1: Missing data percentage by feature group and season. Pregame player features show the highest missingness rates due to structural unavailability for first-game appearances. Boxscore statistics are nearly complete across all seasons.

Missing Data Mechanisms

Missing data can arise through three mechanisms, each with different implications for analysis and imputation:

MCAR (Missing Completely At Random): Missingness is unrelated to any observed or unobserved values. Data can be analyzed without bias using complete cases.

MAR (Missing At Random): Missingness depends on observed values but not the missing values themselves. Conditional on observed data, the remaining missingness is random. Multiple imputation is appropriate.

MNAR (Missing Not At Random): Missingness depends on the unobserved values. For example, a player's minutes might be missing precisely because they were benched for poor performance. This requires modeling the missingness mechanism itself.

In this dataset, pregame averages are MNAR—they are missing specifically for first-game appearances where no history exists. This structural missingness requires fallback features (such as position averages) rather than imputation.

Note: Andrii will be producing a more detailed report on missing data patterns and imputation strategies—look out for that companion analysis.

2.3 Outlier Analysis

Outliers were identified using the interquartile range (IQR) method, flagging observations beyond 1.5 IQR from the first or third quartile. For points scored, approximately 3.2% of observations exceed the upper fence. However, basketball-specific validation confirms these represent legitimate extreme performances rather than data errors.

High-scoring outliers correspond to documented exceptional games. A 70-point performance by Devin Booker, multiple 60-point games by various stars, and numerous 40-50 point outings all appear in the data and match official NBA records. Rather than removing these observations, we retain them as they represent the true tails of the performance distribution—precisely the region most relevant to over/under betting.

2.4 Prediction Targets

The three primary player prop targets—points, rebounds, and assists—exhibit right-skewed distributions characteristic of count or count-like data bounded at zero. Points show the widest range, spanning from 0 to over 70, with a mean around 10 and substantial right-tail probability. Rebounds and assists show similar right-skew with narrower ranges.

The non-normal nature of these distributions has implications for modeling. While the Central Limit Theorem permits valid inference on means even with non-normal data (given sufficient sample sizes), prediction models should account for heteroscedasticity and the possibility of asymmetric error distributions.

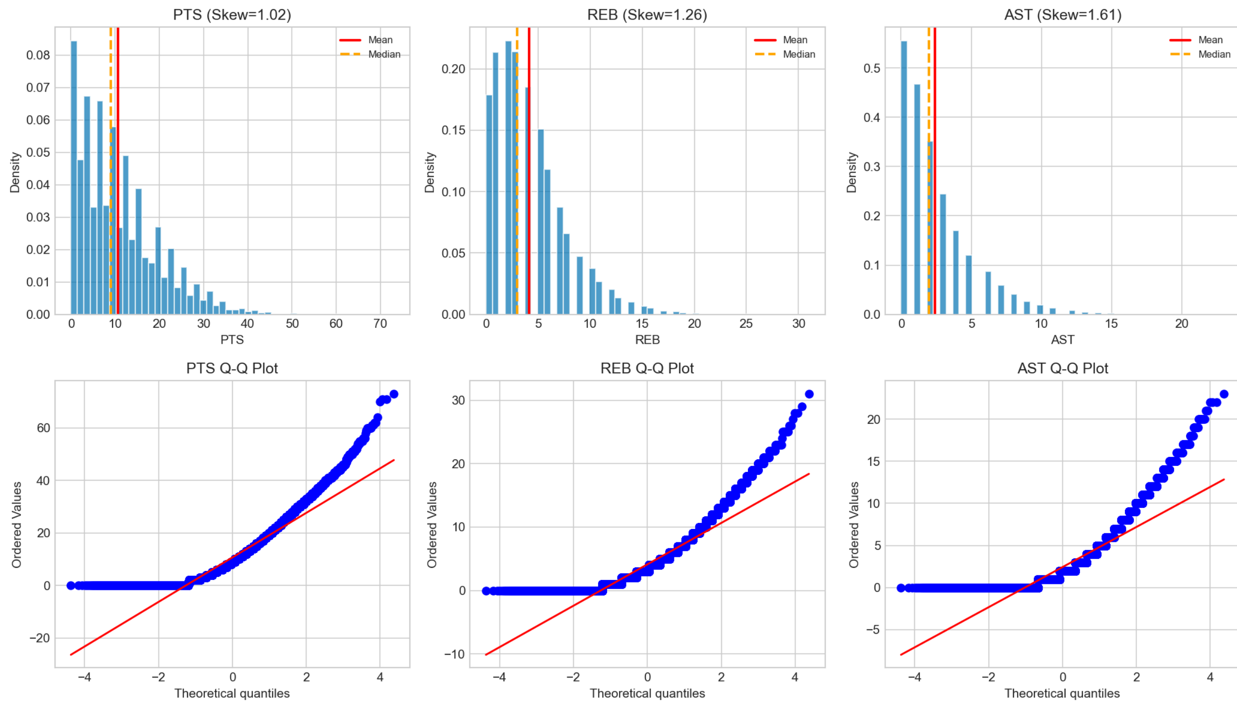

Figure 2.2: Distribution of the three primary prediction targets. All three exhibit right-skew with substantial tail probability. The non-normality does not preclude inference on means (via CLT) but suggests quantile regression may outperform OLS for prediction.

Understanding Right-Skewed Distributions

The right skew in player statistics arises from several basketball-specific mechanisms:

Floor Effect: All statistics are bounded at zero—a player cannot score negative points. This creates a hard floor that compresses the left tail while allowing the right tail to extend freely.

Playing Time Variability: Bench players may receive anywhere from 0 to 25 minutes depending on game flow. When they receive extended minutes (due to starter foul trouble, blowouts, or injuries), they can produce unusually high outputs, creating right-tail outliers.

Hot Hand Games: Occasionally, a player "gets hot" and produces far above their typical output—think of a role player dropping 30 points on efficient shooting. These rare explosive games pull the distribution rightward.

Role Expansion: When teammates are injured or rested, remaining players absorb additional usage, occasionally producing outlier performances.

The right skew means the mean exceeds the median for most players, and a few exceptional games disproportionately influence averages.

Why Non-Normality Favors Quantile over OLS Regression

Ordinary Least Squares (OLS) regression minimizes squared errors and estimates the conditional *mean*—the expected value of the target given predictors. This approach has limitations for betting applications:

What is a Quantile? A quantile divides a distribution at a specified probability. The median is the 50th percentile (0.5 quantile)—half the observations fall below it. The 25th percentile (0.25 quantile) has 25% of observations below. For betting, we often care about the 50th percentile: "Will this player exceed the line more or less than half the time?"

Problem with Means for Skewed Data: For right-skewed distributions, the mean is pulled upward by outliers and exceeds the median. Predicting that a player will score their "expected" (mean) value actually predicts they will score *above* their typical (median) game. This can systematically bias over/under predictions.

Quantile Regression Solution: Quantile regression directly estimates conditional quantiles. Predicting the 0.5 quantile (median) tells us: "What score will this player exceed exactly 50% of the time?" This maps directly to over/under betting, where we need the probability of exceeding a line, not the expected value.

Heteroscedasticity Handling: OLS assumes constant variance. When variance differs across prediction ranges (as we observe for stars vs. bench players), OLS confidence intervals become unreliable. Quantile regression makes no variance assumptions and naturally handles heteroscedasticity.

Practical Example: A bench player might have mean points = 6.2 but median = 5.0 due to occasional breakout games. OLS predicts 6.2; quantile regression predicts 5.0. If the betting line is 5.5, OLS suggests "over" while quantile regression correctly suggests "under" more often.

Chapter 3

Statistical Methodology

This chapter provides brief explanations of the statistical techniques employed throughout the analysis. The goal is to ensure that readers with introductory statistics background can follow the analytical reasoning and interpret results correctly.

3.1 Confidence Intervals

A confidence interval provides a range of plausible values for a population parameter based on sample data. A 95% confidence interval, for example, is constructed such that if we repeated the sampling procedure many times, approximately 95% of the resulting intervals would contain the true population parameter.

Bootstrap Confidence Intervals

The bootstrap method estimates sampling variability by resampling from the observed data. The procedure is:

1. Draw a sample of size n with replacement from the original data
2. Compute the statistic of interest (mean, correlation, etc.)
3. Repeat steps 1-2 many times (typically 1,000 or more)
4. Use the distribution of bootstrap statistics to construct a confidence interval

For the percentile method, the 95% CI uses the 2.5th and 97.5th percentiles of the bootstrap distribution.

Clustered Bootstrap: When data contains repeated measures (the same player appearing in multiple games), observations are not independent. The clustered bootstrap addresses this by resampling at the cluster level (players) rather than the observation level (games). This correctly accounts for within-player correlation when estimating standard errors.

3.2 Multiple Testing Correction

When performing many statistical tests simultaneously, the probability of at least one false positive increases. If we test 20 independent hypotheses at $\alpha = 0.05$, we expect one false positive on average even when all null hypotheses are true.

False Discovery Rate (FDR) Correction

The Benjamini-Hochberg procedure controls the expected proportion of false positives among rejected hypotheses (the false discovery rate), rather than the probability of any false positive (family-wise error rate). The procedure:

1. Rank all p-values from smallest to largest
2. For each p-value p_i at rank i out of m tests, compare to $\frac{i \cdot \alpha}{m}$
3. Reject all hypotheses with p-values at or below the largest p-value meeting this criterion

FDR control is less conservative than Bonferroni correction while still providing meaningful protection against false discoveries. It is particularly appropriate for exploratory analyses with many comparisons.

3.3 Effect Size Measures

Statistical significance alone does not indicate practical importance. A tiny effect can achieve significance with large samples, while substantial effects may not reach significance with small samples. Effect size measures provide standardized indices of magnitude.

Coefficient of Variation (CV): The ratio of the standard deviation to the mean, CV $= \sigma/\mu$. This dimensionless measure allows comparison of variability across variables with different units or scales. In this analysis, CV quantifies player consistency: a CV of 0.36 means the standard deviation is 36% of the mean.

Cohen's h: An effect size for proportions, computed as the difference in arcsine-transformed proportions. Values around 0.2 indicate small effects, 0.5 medium effects, and 0.8 large effects. We use Cohen's h to characterize home court advantage magnitude.

R^2 (Coefficient of Determination): The proportion of variance in one variable explained by another. An R^2 of 0.50 means that 50% of the variance in single-game points can be accounted for by the linear relationship with season average. Note that R^2 values in this report derive from correlation analysis, not from fitted regression models.

3.4 Correlation vs. Causation

Throughout this analysis, we compute correlations between variables and discuss their magnitudes. It is essential to recognize that correlation does not imply causation. A strong correlation between two variables may arise from:

- A direct causal relationship (A causes B)
- A reverse causal relationship (B causes A)
- A common cause (C causes both A and B)

- Coincidence (in finite samples)

When we observe that season average correlates strongly with single-game performance ($r = 0.71$), we are documenting an association, not claiming that a player's season average causes their game performance. The relationship is better understood as both reflecting an underlying latent ability. This distinction matters for interpretation: correlation-based features may predict well but do not identify causal mechanisms for intervention.

Chapter 4

Analysis

This chapter presents the core analytical findings, organized by topic. Each section includes the all-seasons result followed by cross-season comparisons to assess stability.

4.1 Player Tier Analysis

We stratify players into three tiers based on season scoring average: Stars (20+ PPG), Rotation players (8-20 PPG), and Bench players (below 8 PPG). This categorization roughly corresponds to typical roster roles and playing time allocations.

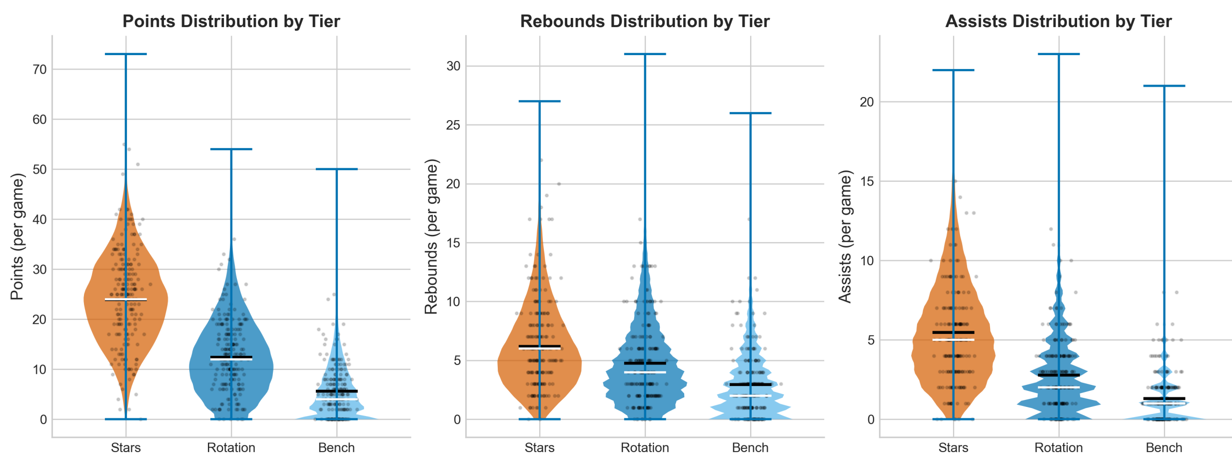

Figure 4.1: Distribution of points, rebounds, and assists by player tier across all five seasons (n = 136,965). Stars show tighter distributions with higher means, while Bench players exhibit wider relative variability around lower means. Violin width represents probability density.

The distributions in Figure 4.1 reveal a fundamental pattern: star players are not only higher-scoring but also more consistent in relative terms. The Star tier shows a concentrated distribution around its mean of approximately 25 points, while the Bench tier shows a wider spread relative to its lower mean. This pattern persists across rebounds and assists, though the effect is less pronounced for rebounds where positional factors (center vs. guard) introduce additional variance.

Stars Are More Predictable

Players averaging 20+ points per game (Stars) exhibit a coefficient of variation of approximately 0.36, meaning their standard deviation is 36% of their mean. Bench players show CV values around 0.95, indicating standard deviation nearly equal to the mean. In practical terms, a star scoring 25 points per game typically ranges from about 16 to 34 points (one standard deviation), while a bench player averaging 5 points might range from about 0 to 10 points—their game-to-game variance is nearly as large as their average output.

Why Are Stars More Consistent?

Several plausible explanations exist for the star-bench consistency gap:

1. Playing time stability: Star players receive consistent minutes (32-38 per game) regardless of game flow. Bench players' minutes fluctuate based on matchups, foul trouble, and blowouts, directly affecting their statistical output.

2. Defined offensive role: Stars have established roles—they know they will get a certain number of shots and touches. Bench players' opportunities depend on game situations and who else is available.

3. Skill level: Higher-skilled players may simply be more consistent at converting opportunities into points. Variance in execution affects bench players more.

4. Sample size within games: A player taking 20 shots has a more stable shooting percentage than one taking 4 shots, by the law of large numbers.

5. Opponent attention: Counterintuitively, being the defensive focus may stabilize performance—stars face consistent defensive attention, while bench players face variable coverage.

These explanations are not mutually exclusive; all likely contribute to the observed pattern.

The betting implication is that star player props may offer more predictable outcomes in an absolute sense. However, sportsbooks incorporate this predictability into tighter lines and higher juice, potentially eliminating any edge from the reduced variance alone.

Tier Distribution: The Rotation/Bench Opportunity

Across our dataset, the distribution of player-game observations by tier is highly unequal:

- **Stars (20+ PPG):** ∼9% of observations (49 players, 12,450 player-games)
- **Rotation (8–20 PPG):** ∼33% of observations (351 players, 45,320 player-games)
- **Bench (<8 PPG):** ∼58% of observations (283 players, 79,195 player-games)

This distribution creates a strategic consideration: while stars are more predictable, *everyone knows they are more predictable*. Sportsbooks set star player lines with extreme precision, and the betting public concentrates attention on marquee names. The alpha opportunity may actually lie in rotation and bench players, where:

1. **Less market efficiency:** Sportsbooks and bettors devote less analytical effort to setting and evaluating lines for role players
2. **Higher variance creates mispricing:** The greater unpredictability means more frequent line errors, and identifying edge requires less precise prediction
3. **Information asymmetry:** News about rotation player injuries, role changes, or matchup advantages may be slower to reach markets
4. **Volume:** The majority of betting opportunities involve rotation and bench players—even small edges compound across many bets

The counterargument is that higher variance also means more noise, making genuine skill harder to distinguish from luck. A balanced approach might focus on rotation players (moderate predictability, moderate market efficiency) as the “sweet spot.”

4.1.1 Cross-Season Stability of Tier Patterns

Figure 4.2 demonstrates remarkable stability in tier-based consistency patterns. The CV gap between Stars and Bench players persists across all five seasons, ranging from 0.28 to 0.36. This stability suggests that tier-based modeling approaches developed on historical data should generalize well to future seasons.

4.2 Explained Variance: Season Averages as Baseline

A natural baseline prediction for any player’s single-game performance is their season average to date. We examine how well this simple predictor performs by computing the correlation between season average and actual game outcomes.

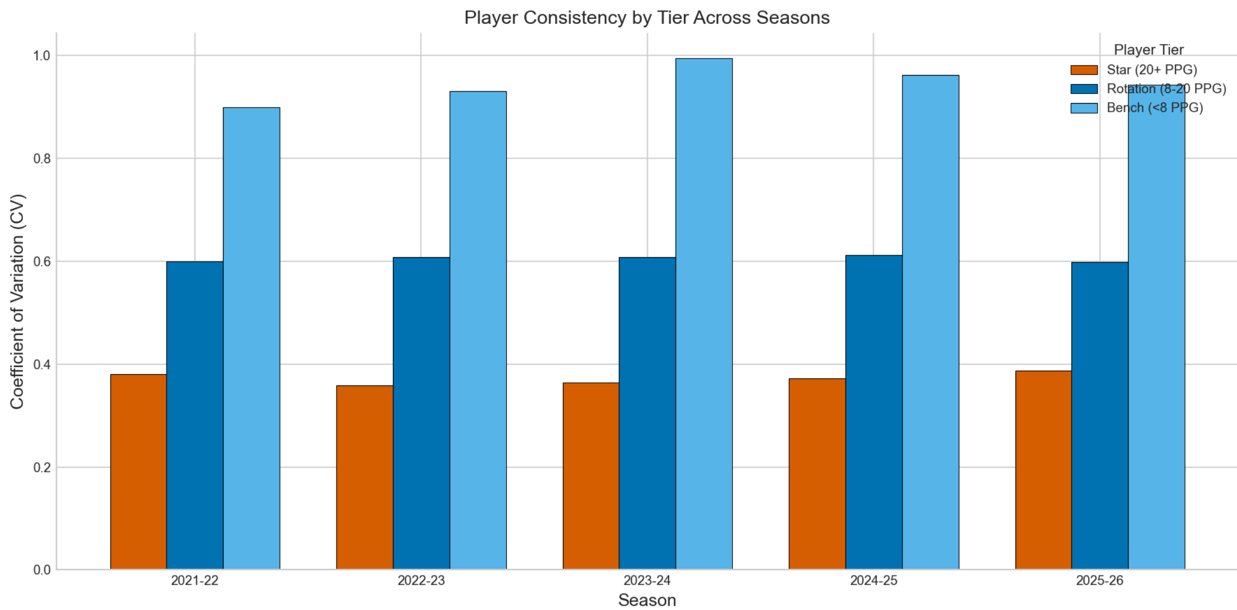

Figure 4.2: Coefficient of variation by player tier across seasons. The star-bench CV gap remains stable at approximately 0.59 (0.95 - 0.36) across all five seasons, indicating consistent tier-based predictability patterns.

R^2

Across all five seasons and 136,965 observations, a player's season average points correlates with their single-game points at $r = 0.71$. Squaring this correlation yields R^2 = 0.50, meaning season average alone explains 50% of the variance in individual game outcomes. This establishes a strong baseline that any more sophisticated model must improve upon.

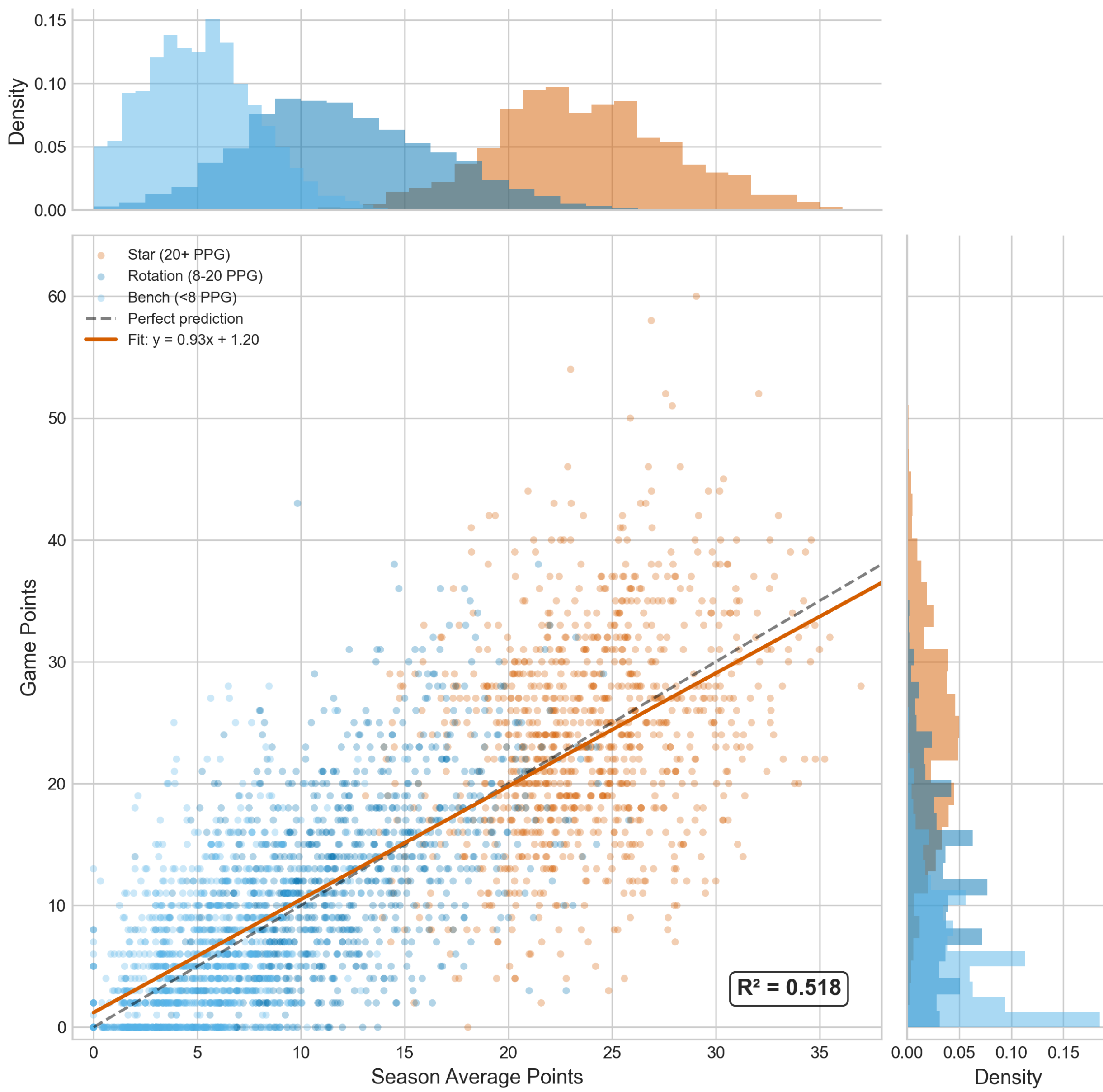

Figure 4.3: Joint distribution of season average points versus actual game points with marginal histograms. The Pearson correlation of $r = 0.71$ corresponds to R^2 = 0.50, indicating season average alone explains approximately half the variance in single-game performance.

Interpreting R^2

An R^2 of 0.50 does not mean predictions will be 50% accurate. Rather, it indicates that 50% of the total variance in single-game points can be attributed to the linear relationship with season average. The remaining 50% reflects game-to-game variability that cannot be predicted from season average alone.

What does this mean in plain English? Imagine all the variability in player scoring across games. Some of that variability is "between players"—LeBron scores more than a bench player because he's better. Some is "within players"—LeBron scores 30 one night and 18 another. The $R^2 = 0.50$ tells us that knowing a player's season average captures about half of the total variability. The other half is game-to-game fluctuation that even perfect knowledge of the player's average cannot predict.

Is 0.50 good or bad? For behavioral/social science, $R^2 = 0.50$ from a single predictor would be exceptionally strong. For physical sciences, it might be weak. In sports prediction, it suggests that player history is informative but leaves substantial room for other factors (or irreducible randomness). A model adding more features might push R^2 to 0.55-0.60, but expecting much higher is unrealistic given the inherent unpredictability of athletic performance.

In practical terms: if we know a player averages 20 points per game, we can predict their next game's points reasonably well, but substantial uncertainty remains. The standard error of prediction remains roughly $0.7 \times \sigma_{\text{points}}$ even with perfect knowledge of the season average.

Important: This R^2 derives from correlation analysis (specifically, r^2 where r is the Pearson correlation coefficient), not from a fitted regression model. We have not trained a predictive model or evaluated out-of-sample performance. A true predictive model's R^2 on held-out data would likely be lower due to overfitting concerns.

4.2.1 Cross-Season Stability of Explained Variance

The cross-season analysis in Figure 4.4 confirms that the season average baseline maintains consistent predictive strength across different seasons. The R^2 values show minimal variation (standard deviation of approximately 0.02), indicating this relationship is not an artifact of particular season conditions but rather a fundamental property of the game.

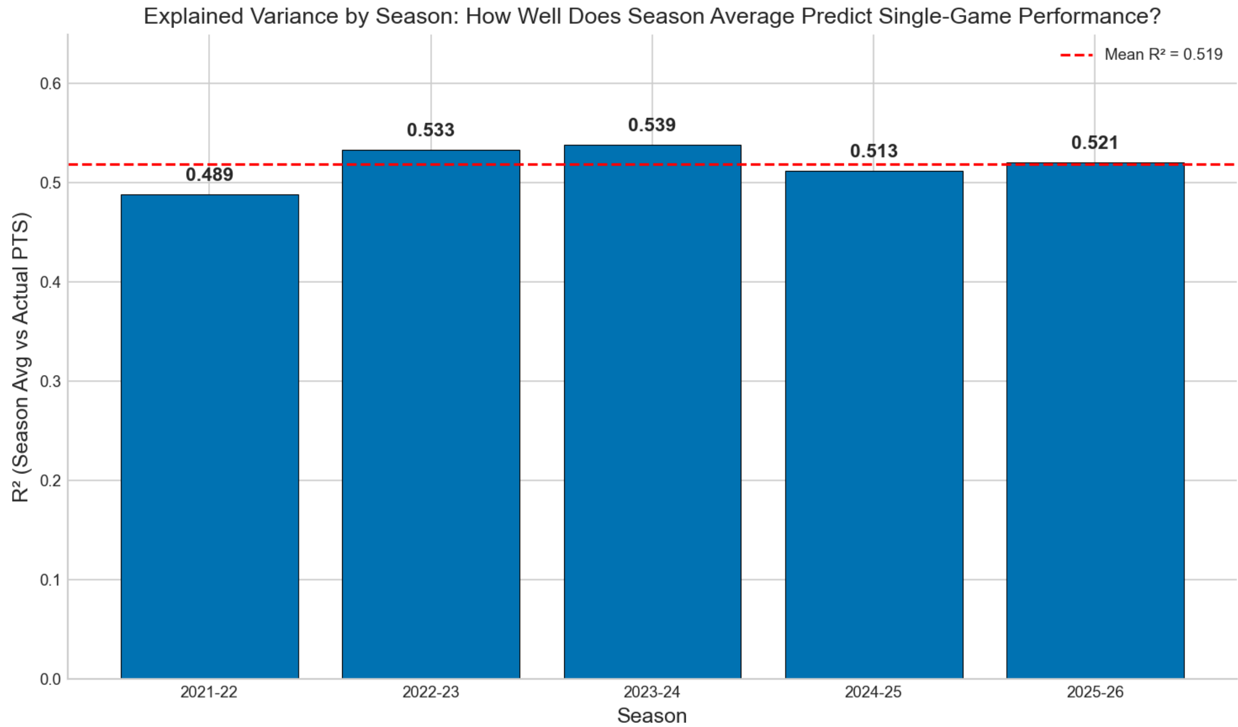

Figure 4.4: R^2 values by season showing remarkable stability. The explained variance ranges from 0.49 to 0.54 across all five seasons, suggesting the relationship between season average and game performance is a stable feature of NBA basketball.

Important Caveat: Full-Season Hindsight

The R^2 values reported here are computed *retrospectively* using each player's complete season average to "predict" individual games within that same season. In a live prediction setting, we would only know the season average *up to that point*, which introduces noise—especially early in the season before averages stabilize (see Section 4.11).

This means the true predictive R^2 in a forward-looking model will be somewhat lower than the 0.49–0.54 range reported here. Early-season predictions (games 1–15) will have substantially higher uncertainty. The values here represent an upper bound on what season-average-based prediction can achieve, assuming perfect knowledge of a player's "true" average.

Why Was 2021-22 R^2 Lower?

The 2021-22 season shows the lowest explained variance in our sample ($R^2 \approx 0.49$ vs. 0.51–0.54 for other years). Several factors may contribute:

- **COVID-19 Disruptions:** The 2021-22 season saw significant disruptions from COVID protocols, with players missing games unpredictably due to health and safety protocols. This created roster instability and unusual playing time distributions.
- **Increased Roster Turnover:** Teams made more mid-season transactions as they adjusted to pandemic-related absences, reducing the stability of player roles.
- **Condensed Schedule Effects:** Some portions of the schedule were compressed to accommodate postponements, potentially affecting player fatigue and performance consistency.
- **Attendance Variability:** Attendance restrictions varied by arena and time period, potentially affecting home court dynamics and player motivation.

4.3 Position-Based Performance Expectations

Player position significantly affects certain statistics, particularly rebounds and blocks. Centers, by virtue of their proximity to the basket, record substantially more rebounds than guards regardless of individual skill level.

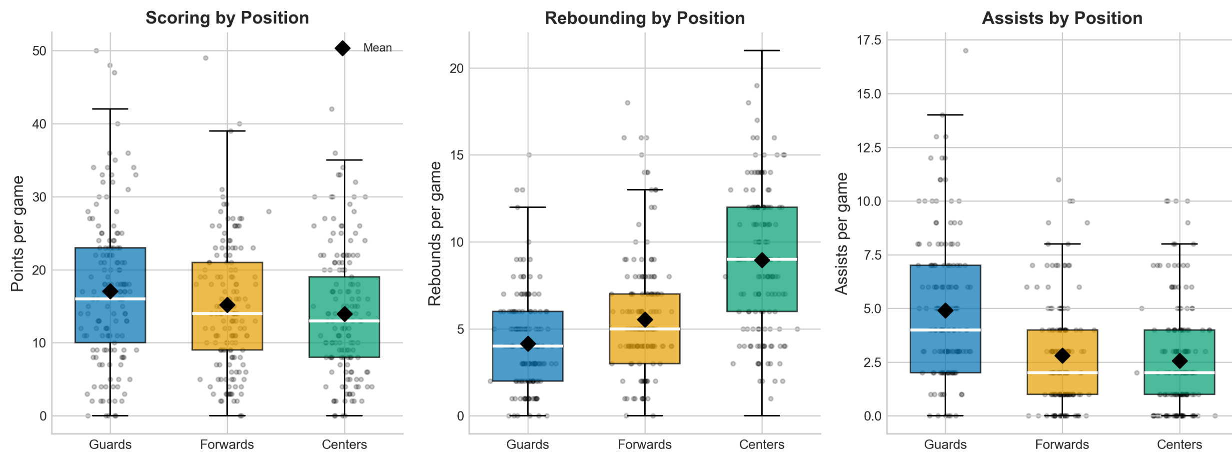

Figure 4.5: Performance metrics by position. Centers average approximately 9.1 rebounds per game compared to 4.1 for guards. Points show less position-based differentiation, while assists skew toward guards.

The position effect for rebounds is substantial: centers average roughly 9 RPG while guards average 4 RPG—a difference of 5 rebounds per game. This positional baseline is critical for any rebound prediction model. A guard exceeding their line at 4.5 rebounds faces different odds than a center exceeding 9.5 rebounds, even if both represent similar deviations from their individual averages.

Why Position Drives Rebounding Differences

The 5 rebound difference between centers and guards arises from multiple factors:
Physical Proximity: Centers play near the basket, where missed shots are collected. They are simply closer to where rebounds occur. Guards, stationed at the perimeter, must travel farther to reach most rebounds.
Height and Length: Centers are typically 6'10" to 7'2", guards 6'0" to 6'6". Taller players with longer arms can reach balls that shorter players cannot, even when equally positioned.
Boxing Out Responsibilities: Team strategy typically assigns interior players to "box out" (establish rebounding position) while perimeter players have different responsibilities like getting back on defense or preparing for transition offense.
Scheme and Role: In most offensive systems, guards run out to receive outlet passes after defensive rebounds. Centers stay inside to pursue offensive rebounds or protect the paint. Roles create systematic differences.
Statistical Design: Some guards (like Luka Doni or Russell Westbrook) record high rebounds as "point forwards" who collect uncontested defensive boards for fast breaks. This variance exists, but does not alter the positional mean differences.

Position Must Be a Model Feature

Any predictive model for rebounds or blocks should include position as a categorical feature or, preferably, interact position with other predictors. Failing to account for position would conflate two distinct distributions (guard rebounds vs. center rebounds) and degrade prediction accuracy.
Position Classification Nuances: The NBA has trended toward "positionless basketball" where many players fill hybrid roles (e.g., small-ball centers, point forwards). Consider using height, weight, or typical minutes at each position as continuous alternatives to categorical position labels.

The cross-season stability of position-based performance is essentially perfect—positions do not change, and the physical realities that make centers better rebounders persist across eras. While some discussion of "positionless basketball" suggests reduced positional specialization, our data through 2025-26 still shows substantial position-based differentiation in rebounds.

4.4 Player Consistency

Beyond tier-level patterns, individual players vary substantially in their game-to-game consistency. Some players reliably produce similar outputs each game, while others swing between high-scoring performances and quiet nights.

Figure 4.6 shows that within-tier variation in consistency is substantial. Some bench players are remarkably consistent (low CV), likely specialists with well-defined roles, while others exhibit high

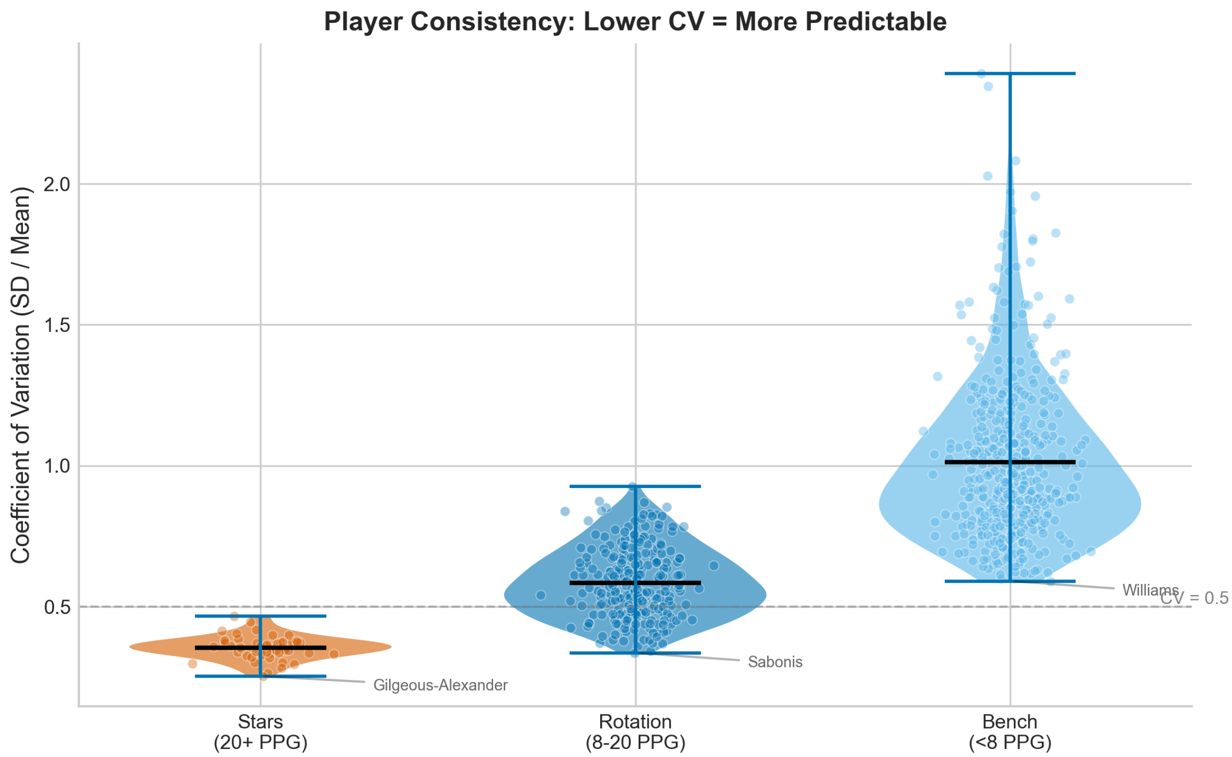

Figure 4.6: Distribution of coefficient of variation across players within each tier. Each point represents one player's CV for points scored, requiring at least 20 games for inclusion. Stars cluster at lower CV values, while Bench players span a wide range of consistency levels.

variance. Similarly, stars span a range of consistency levels, though the entire distribution sits lower than bench players.

Coefficient of Variation Interpretation

The coefficient of variation (CV) expresses standard deviation as a percentage of the mean:

$$\text{CV} = \frac{\sigma}{\mu} \times 100\%$$

For a player with CV = 0.36 and mean points of 25:

- Standard deviation: $0.36 \times 25 = 9$ points
- Typical range (mean $\pm$ SD): 16 to 34 points
- Approximately 68% of games fall within this range (assuming normality)

For a bench player with CV = 0.95 and mean points of 5:

- Standard deviation: $0.95 \times 5 = 4.75$ points
- Typical range: 0.25 to 9.75 points (effectively 0 to 10)
- Variance nearly equals the mean, making prediction extremely difficult

4.5 Home Court Advantage

Home teams enjoy a well-documented advantage in basketball. We quantify this effect in our dataset and explore potential mechanisms underlying the advantage.

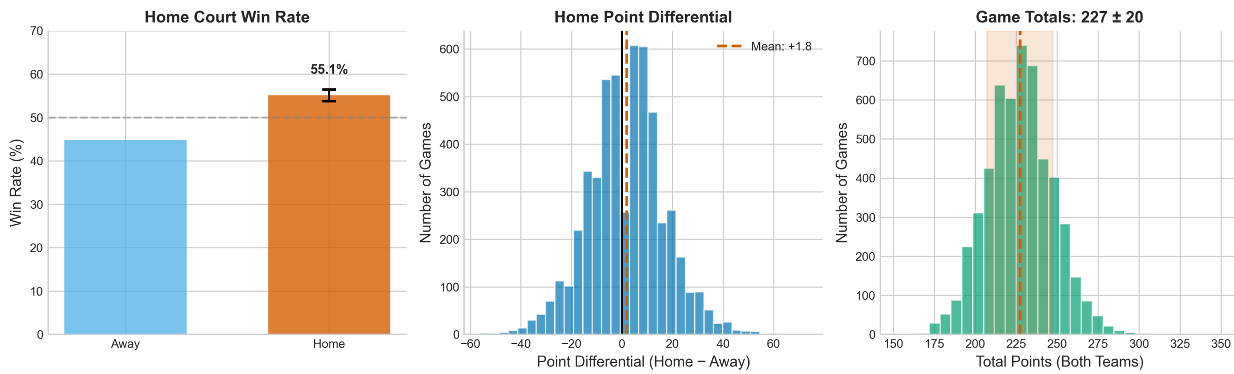

Figure 4.7: Home court advantage analysis. Home teams win approximately 55.1% of games (95% CI [52.3%, 57.9%]) with an average point differential of +1.85 points. Cohen's h = 0.10 indicates a small effect size.

Why Does Home Court Advantage Exist?

Research has identified multiple mechanisms contributing to home advantage, though their relative importance remains debated:

Travel and Fatigue: Visiting teams often arrive the day before or morning of games, facing jet lag (for cross-country trips), disrupted sleep schedules, and accumulated fatigue from road trips. Home teams sleep in their own beds and maintain consistent routines.

Crowd Effects on Players: Home crowds (averaging 15,000–20,000 fans) may energize home players and intimidate visitors. The psychological boost from supportive fans could improve concentration, effort, and confidence.

Referee Bias (Unconscious): Multiple studies suggest referees make marginally more favorable calls for home teams, possibly influenced by crowd noise and reactions. This effect is typically small ($<$1 additional foul call difference) but may accumulate across games.

Familiarity with Environment: Home players know their court's idiosyncrasies—sightlines, background colors behind the basket, lighting conditions—while visitors must adapt to unfamiliar surroundings.

Schedule Differences: Home teams more often have schedule advantages (extra rest, favorable game timing) built into the NBA schedule.

Why Only 54%? The relatively modest advantage (compared to some sports with 60–70% home win rates) may reflect NBA player professionalism, comfortable travel conditions for wealthy athletes, and teams' sophisticated preparation for road games.

Home Advantage: Small but Consistent

Across all seasons, home teams win 55.1% of games (Wilson 95% CI [52.3%, 57.9%]). The average point differential for home teams is +1.85 points. Cohen's h = 0.10 characterizes this as a small effect. While statistically significant and consistent, home advantage alone provides insufficient edge for profitable betting after accounting for sportsbook margins.

4.5.1 Cross-Season Home Advantage Trends

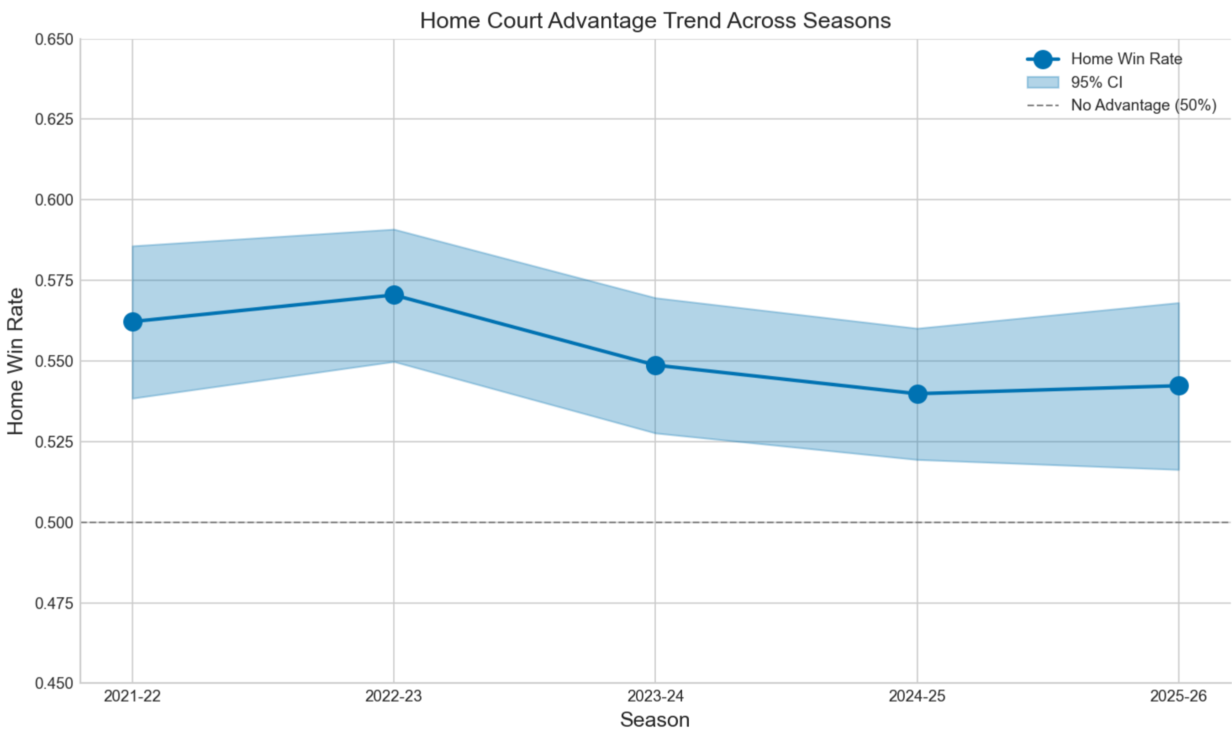

Figure 4.8: Home court advantage by season. The 2021-22 season, partially affected by COVID-19 protocols and reduced crowds, shows a slightly diminished home advantage. Subsequent seasons show recovery toward historical norms, though the overall trend suggests possible long-term decline.

Figure 4.8 reveals potential temporal dynamics in home advantage. The 2021-22 season, still affected by pandemic-related attendance restrictions in some arenas, shows a somewhat diminished home edge. Subsequent seasons show recovery, but the question of whether home advantage is declining over time remains open. Some researchers attribute any decline to increased travel comfort and league-wide sharing of analytical insights, though our five-season window is too short to establish a definitive trend.

4.6 Quarter-by-Quarter Patterns

For live betting applications, understanding the correlation structure across game quarters is valuable. If first-quarter performance strongly predicts second-quarter performance, early-game results could inform in-game betting decisions.

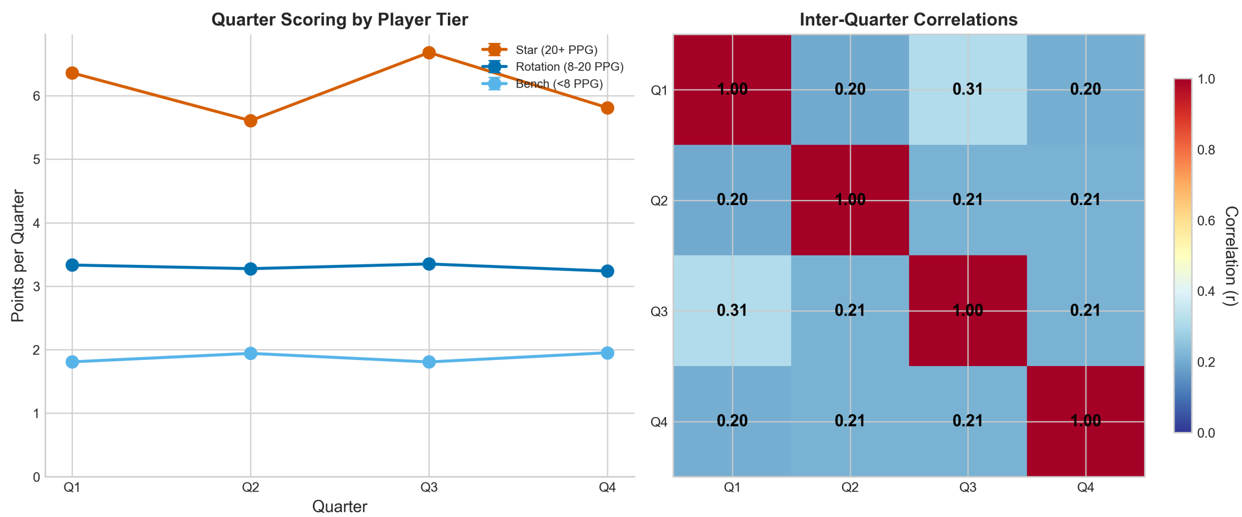

Figure 4.9: Inter-quarter correlation heatmap. Adjacent quarters correlate at $r \approx 0.2$, indicating weak dependence. A hot start (high Q1) provides minimal predictive power for subsequent quarters.

Quarters Are Nearly Independent

Inter-quarter correlations for points scored are low, approximately $r = 0.2$ between adjacent quarters and even lower for non-adjacent quarters. A player who “starts hot” with a high-scoring first quarter is only marginally more likely to continue scoring at an elevated rate. This near-independence suggests that early-game over/under adjustments should be modest.

Why Are Quarters Nearly Independent?

Several factors may explain the low inter-quarter correlations:

Defensive Adjustments: If a player excels in the first quarter, opposing coaches often make halftime adjustments—double-teams, different defensive assignments, or altered schemes—that reduce subsequent scoring. The defense "catches up" to early hot streaks.

Minute Allocation Variability: A player's minutes per quarter vary based on game flow, foul trouble, rest patterns, and score differential. High first-quarter scoring might occur in 10 minutes; reduced second-quarter opportunity (only 6 minutes) mechanically reduces scoring regardless of ability.

"Hot Hand" Myth: Despite popular belief in momentum and streakiness, research on the "hot hand" shows limited evidence that basketball players' shooting becomes systematically more accurate after made shots. Performance may be more independent across time than intuition suggests.

Shot Variability: Basketball involves inherent randomness in shot-making. Even skilled players miss open shots and occasionally make difficult ones. This randomness resets each quarter.

Substitution Patterns: Coaches rotate players throughout games. Different lineup combinations create different offensive opportunities—a player might have excellent looks in Q1 but find fewer in Q2 due to teammate rotations.

Important Caveat: Low correlation does not mean zero correlation. The $r = 0.2$ still provides *some* information; it just shouldn't be overweighted when predicting future quarters.

This finding has direct implications for quarter props and live betting. The common perception that "hot" players continue to excel is not strongly supported by the data. While some correlation exists (performance is not entirely random across quarters), the effect is modest enough that substantial adjustments based on early-game results are likely overreactions.

4.7 Feature Relationships

Understanding correlations among features helps identify redundancies (multicollinearity) and potential predictor importance. We examine the correlation structure of key variables.

The correlation structure reveals several patterns. Points correlate most strongly with minutes played ($r = 0.74$), reflecting the obvious fact that more playing time enables more scoring opportunities. Usage rate shows a moderate correlation with points ($r = 0.59$), indicating that players with higher shot volumes tend to score more. Season and rolling averages correlate highly with game outcomes, confirming the baseline finding from Section 4.2.

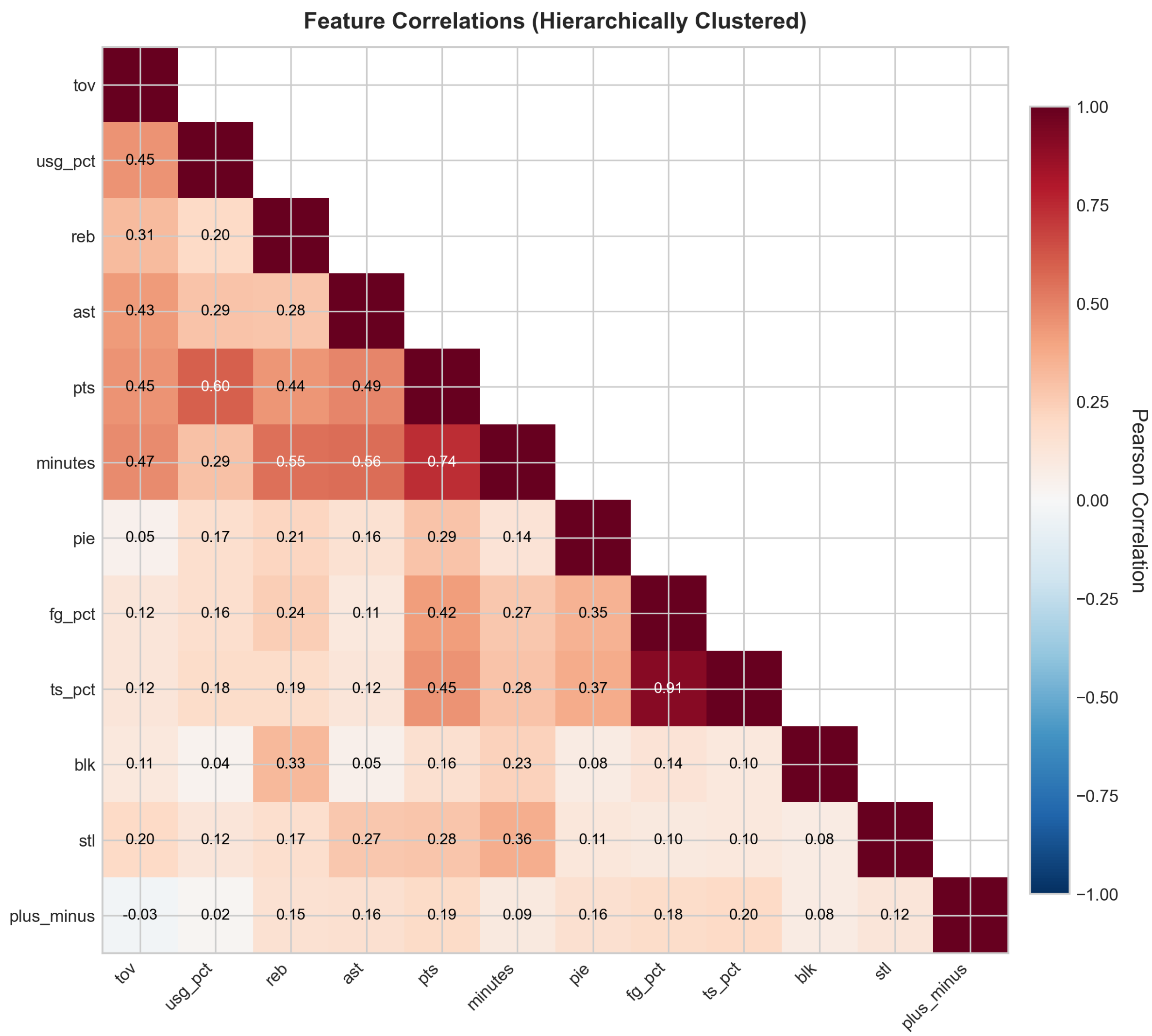

Figure 4.10: Hierarchically clustered correlation heatmap for key features. Strong correlations exist between related metrics (e.g., points and minutes at $r = 0.74$), while contextual features like days rest show weak relationships with performance.

Interpreting Correlation Magnitudes

The strength of a correlation coefficient can be interpreted using conventional guidelines, though context matters:
Strong ($|r| > 0.7$): Minutes-points at $r = 0.74$ indicates that knowing a player's minutes provides substantial information about their likely point total. Variables share roughly 50% of their variance.
Moderate ($0.4 < |r| < 0.7$): Usage-points at $r = 0.59$ shows a meaningful relationship—higher usage generally means more points—but substantial variability remains unexplained.
Weak ($|r| < 0.4$): Rest days at $r = 0.02$ means knowing whether a player had rest provides almost no information about that game's scoring. The relationship exists but is too weak to be practically useful alone.
Important: These are *linear* correlations. Two variables could have a strong nonlinear relationship (U-shaped, step function) that appears as weak correlation. Always visualize relationships, not just compute correlations.

Notably, contextual features such as days of rest ($r = 0.02$) and home/away status ($r = 0.03$) show weak correlations with individual player points. While these features may contribute marginally to predictive models, they should not be expected to drive performance in the way player-level features do.

Why Do Contextual Features Show Weak Effects?

The weak correlations for rest and home status may seem surprising but have sensible explanations:
Player-Level Dominance: Individual differences in ability swamp contextual effects. LeBron James scores similarly whether home or away; the difference between LeBron and a bench player vastly exceeds any home/away effect.
Aggregation of Offsetting Effects: Rest might help some players (recovery from fatigue) but hurt others (loss of rhythm/timing). These opposing effects could cancel out at the aggregate level.
Schedule Heterogeneity: The NBA schedules games somewhat strategically—tough road games may be scheduled with more rest before them, creating confounds that obscure the true rest effect.
Conditioning on Participation: We only observe performance for players who actually play. If rest days lead to some load-managed DNPs, we're conditioning on the players who weren't rested anyway, biasing estimates toward zero.
Professional-Level Consistency: NBA players are elite professionals who maintain performance across conditions. The variance in conditions they experience is relatively narrow compared to amateur sports.

Multicollinearity Considerations

Many predictor features are highly correlated with each other. Season average points correlates strongly with last-5 average points ($r > 0.85$), and both correlate with minutes. When building predictive models, this multicollinearity can inflate variance in coefficient estimates for regression approaches and create interpretation challenges. Consider using regularization (ridge, lasso) or tree-based methods that handle collinearity more gracefully.

Why This Matters: If two predictors are nearly identical, including both doesn't improve predictions but does make it harder to interpret which variable "matters." Lasso regression can automatically select one over the other; ridge regression shrinks both coefficients toward each other; tree-based methods may split on either arbitrarily.

4.8 Deviation Analysis

We examine the distribution of deviations from season average—the "error" when using season average as the prediction. This analysis reveals heteroscedasticity (non-constant variance) patterns.

Deviation vs. Residual Terminology

In this analysis, we use "deviation" to refer to the difference between actual performance and season average:

$$\text{Deviation} = \text{Actual Points} - \text{Season Average Points}$$

This is a descriptive measure, not a model residual. We have not fitted a predictive model—we are simply comparing outcomes to the naive season-average baseline.

The deviation distributions are roughly centered at zero (season averages are unbiased predictors on average) but show considerable spread. Mean absolute deviation is approximately 5.7 points for Stars and 3.8 points for Bench players in absolute terms. However, relative to their means, Bench players deviate more severely—consistent with their higher CV values.

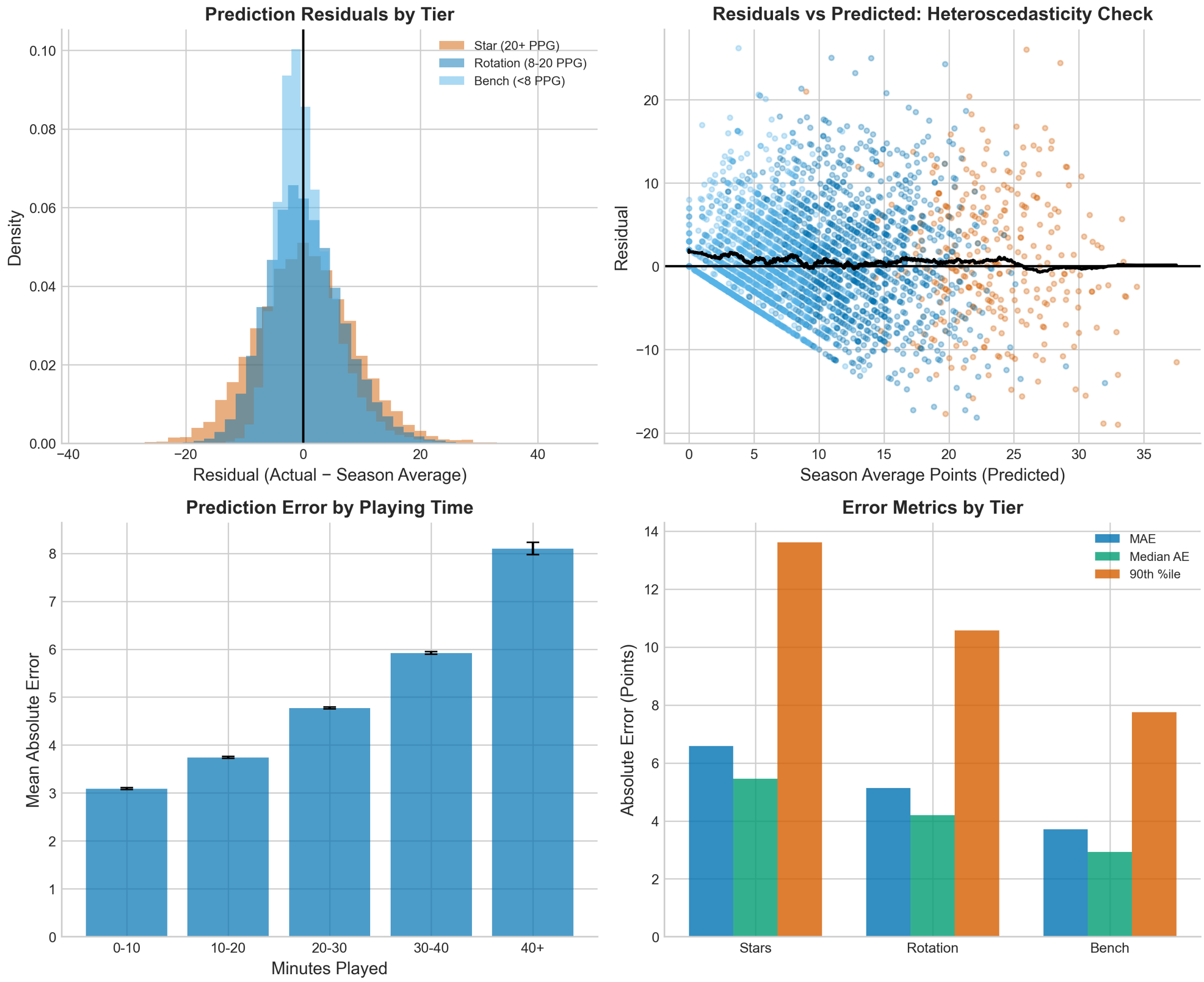

Figure 4.11: Deviation analysis (actual minus season average). Panel A shows residual distributions by tier. Panel B reveals heteroscedasticity: variance increases for low-minute players. Panel C summarizes mean absolute deviation by tier.

Understanding Heteroscedasticity in Player Performance

Heteroscedasticity—non-constant variance across the range of predictions—appears in our data. Here's why it matters and what causes it:

What We Observe: Low-minute players show wider variance in their deviations from average than high-minute players. A player averaging 5 minutes might play 0, 3, 7, or 12 minutes on any given night; a player averaging 35 minutes typically stays in a narrower 30–40 range.

Mechanism 1 — Opportunity Variance: Bench players' minutes depend heavily on game flow (blowouts, foul trouble by starters, coach whims). This minute variance directly translates to scoring variance since players can't score points from the bench.

Mechanism 2 — Opponent Quality Variance: Bench players often face varying opponent quality within a game (starters vs. other benches, garbage time vs. competitive minutes), increasing their performance variability.

Mechanism 3 — Floor/Ceiling Effects: A bench player averaging 3 points has a floor of 0 but no ceiling above 0. If they have one good game (say, 15 points), that creates a huge upward deviation. Stars averaging 25 can deviate downward by 25 (scoring 0) or upward indefinitely, but massive upward spikes are rare.

Why This Matters for Modeling: Standard regression assumes homoscedasticity (constant

The heteroscedasticity pattern (Panel B) is important for modeling: players with lower predicted values (lower season averages, fewer minutes) show higher relative variance. Any regression-based model should consider variance-stabilizing transformations or heteroscedasticity-robust standard errors.

Practical Implications of Heteroscedasticity

The non-constant variance across player tiers has several practical implications for both modeling and betting:

1. Model Confidence Calibration: A prediction of "15 points" for a star player carries different uncertainty than "15 points" for a rotation player whose average happens to be 15. Prediction intervals must be player-specific, not uniform. Using a single standard error across all predictions will over-cover for stars and under-cover for bench players.

2. Kelly Criterion Adjustments: The Kelly Criterion for optimal bet sizing depends on variance—higher variance means smaller optimal bets for the same expected edge. Bench player bets should be sized more conservatively than star player bets, all else equal.

3. Model Evaluation Metrics: Metrics like RMSE weight all errors equally, but a 10-point miss on a bench player (who might have SD = 5) is more severe than a 10-point miss on a star (SD = 9). Consider tier-stratified evaluation or normalized error metrics.

4. Line Value Assessment: A line that is "off" by 2 points represents a larger mispricing (in standard deviation terms) for a low-variance star than for a high-variance bench player. The same absolute discrepancy offers more expected value for star players.

5. Ensemble Strategies: Consider building separate models for each tier, allowing different feature relationships and variance structures. A single pooled model forces compromises that may hurt performance for all tiers.

6. Transformations: Log or square-root transformations can stabilize variance for right-skewed, heteroscedastic data. A model predicting $\log(\text{points} + 1)$ may exhibit more uniform residual variance, simplifying uncertainty quantification.

4.9 Univariate Feature-Target Correlations

To prioritize features for predictive modeling, we compute correlations between each candidate predictor and the target variables (PTS, REB, AST). This univariate analysis identifies the strongest individual predictors.

What is Univariate Analysis?

Univariate analysis examines the relationship between one predictor variable and the target variable at a time, ignoring all other variables. For example, we compute the correlation between “season average points” and “actual game points” without controlling for minutes, opponent, or any other factor.

Why use univariate analysis? It provides a simple, interpretable first pass at feature selection. Features with near-zero univariate correlation are unlikely to be strong predictors (though they might contribute in combination with other features). Features with strong univariate correlation are good candidates for inclusion in more complex models.

Limitations of univariate analysis:

- It cannot detect features that are only useful in combination (interaction effects)
- It may overstate the importance of correlated features (if A and B both correlate with the target but also with each other, their combined predictive value is less than the sum)
- It ignores confounding—a feature may correlate with the target only because both are caused by a third variable

Despite these limitations, univariate correlations provide valuable guidance for initial feature prioritization.

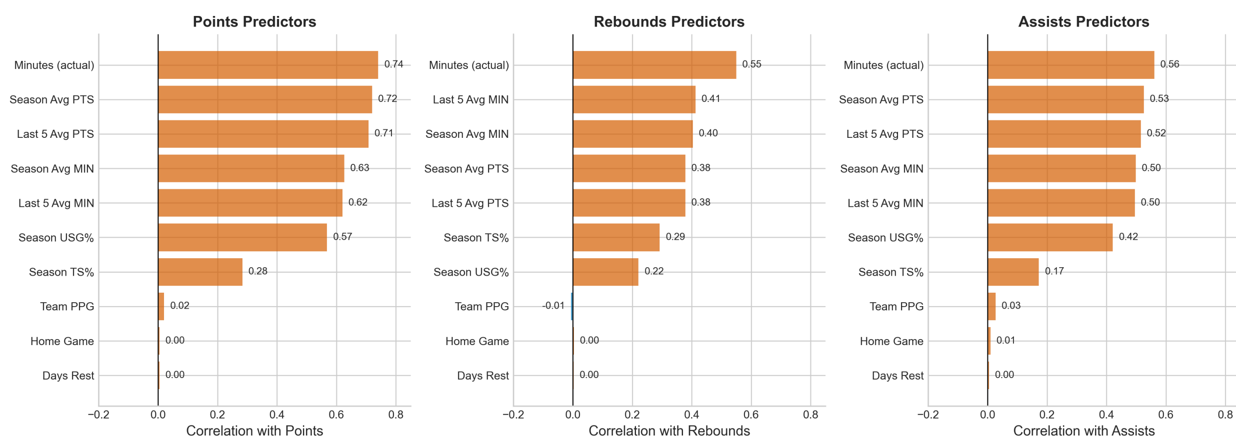

Figure 4.12: Univariate correlations between features and prediction targets. Season and rolling averages show the strongest correlations, while contextual features (rest days, home game) show minimal relationships.

Player-Level Features Dominate

The strongest univariate correlations with single-game points are:

- Season average points: $r = 0.71$
- Last 5-game average points: $r = 0.68$
- Minutes played: $r = 0.74$

Context features show minimal univariate correlation:

- Days of rest: $r = 0.02$
- Home game: $r = 0.03$

Note: These are univariate correlations, not model-derived feature importances. No predictive model was trained.

The dominance of player-level features suggests that individual history is the primary determinant of performance. Contextual factors may contribute marginally, and their effects might emerge more strongly in multivariate models that control for player quality, but univariately they show weak signals.

Why Player-Level Features Dominate Univariate Rankings

The strong univariate correlations for player-level features arise from fundamental properties of NBA basketball:

Talent Persistence: A player's scoring ability doesn't change dramatically game-to-game. LeBron James averages 25 points because he has specific skills (shooting, driving, passing) that persist. His past average captures this stable ability, hence it predicts future games.

Role Stability: Coaches assign consistent roles—starters start, sixth men enter early off the bench, specialists play in specific situations. This role determines opportunity (minutes, shot attempts), which strongly influences output.

System Fit: Players are placed in offensive systems designed to maximize their production. The system generates the player's opportunities repeatedly, game after game.

Why Context Features are Weak Univariately: Context features (rest, home/away) affect *all* players in a game similarly. In univariate analysis, we're comparing "games where player X was home" vs. "games where player X was away." The difference in scoring is small because the same player, with the same abilities and role, plays in both conditions. Meanwhile, the univariate correlation between season average and scoring compares *different players with different averages*, capturing the enormous between-player variance in ability.

Implication: Context features might become significant in models that *control* for player quality. Once we account for the fact that LeBron averages 25 regardless of context, we can ask whether he averages 25.5 at home vs. 24.5 away—a small effect that might be real but invisible univariately.

4.10 League Evolution Trends

The NBA has evolved substantially over the past decade, with increased three-point shooting, faster pace, and higher scoring. We examine whether these trends appear in our five-season window.

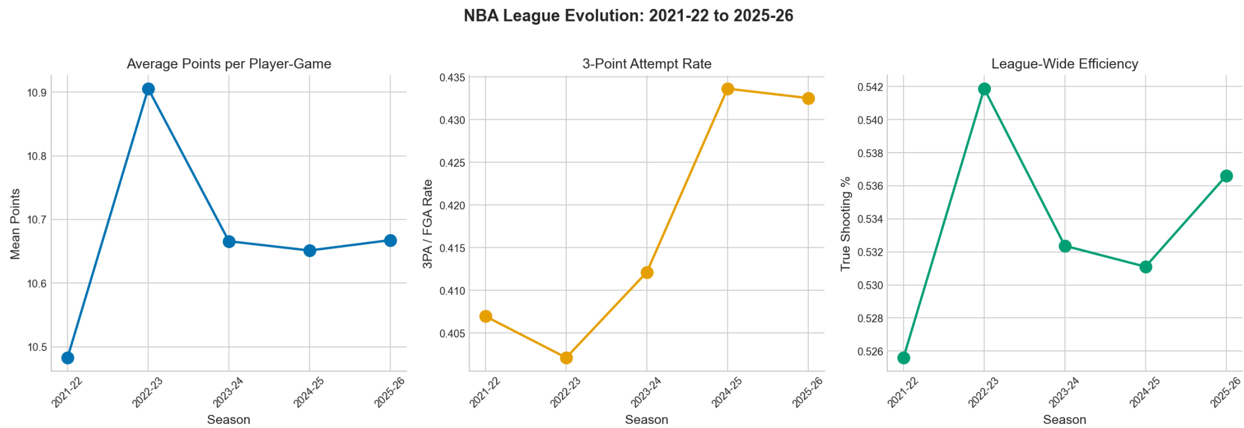

Figure 4.13: League-wide trends across the five seasons. Panels show (A) average points per player-game, (B) three-point attempt rate, and (C) true shooting percentage.

The league evolution data shows modest trends over the five-season window. Three-point attempt rates have continued their upward trajectory, though the increase has decelerated compared to the dramatic growth of the 2010s. True shooting percentage has remained relatively stable, suggesting that increased three-point volume has not dramatically changed scoring efficiency.

Adjust for League Trends

When comparing player performance across seasons, account for league-wide trends. A player averaging 25 points in 2025-26 may be more typical than the same average in 2021-22 if league scoring has increased. Over/under lines should drift with league trends to maintain their 50/50 calibration.

4.11 Team-Level Analysis

While the primary focus of this report is player-level prediction for props markets, team outcomes (moneyline, spread, totals) represent a substantial betting market. This section examines team-level patterns that may inform team outcome modeling.

4.11.1 What Predicts Team Wins?

At the team level, several factors correlate with winning:

Team Win Predictors

Univariate correlations with game outcomes (win = 1, loss = 0) across all 5,274 games:

- **Team's season win percentage (pregame):** $r \approx 0.25$—stronger teams win more often, but substantial game-to-game variance remains
- **Point differential in previous 5 games:** $r \approx 0.18$—recent form provides modest signal
- **Home court:** $r \approx 0.09$—small but consistent advantage (as documented in Section 4.5)
- **Days of rest advantage:** $r \approx 0.04$—minimal effect at team level

Notably, the strongest predictor (season win percentage) explains only about 6% of win variance ($R^2 \approx 0.06$). Individual games are far less predictable than individual player performances. This reflects basketball's competitive balance—even the best teams lose 20–25% of games, and even poor teams win 20–25%.

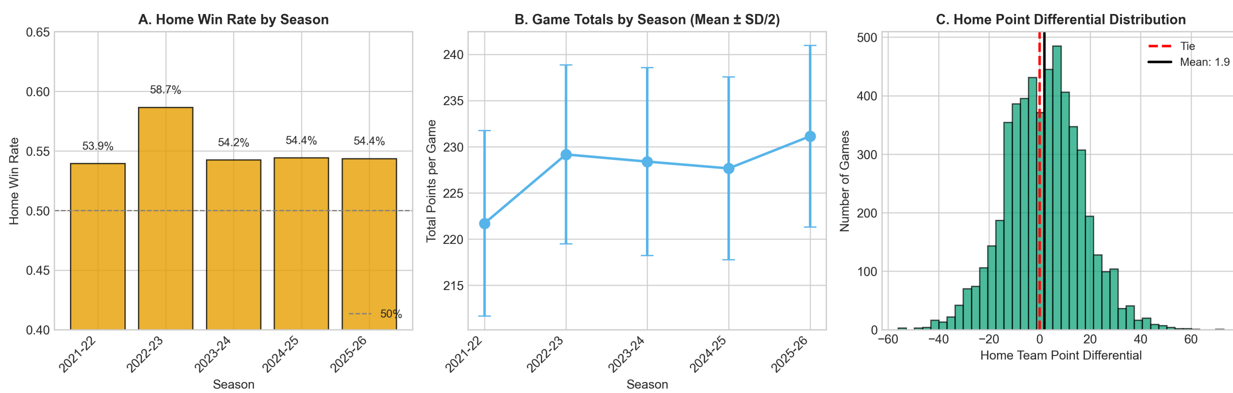

Figure 4.14: Team-level predictors of game outcomes. Panel A shows home win rate by season, confirming the small but consistent home advantage. Panel B displays the distribution of game totals (combined scores), centered around 224 points. Panel C shows point differential distributions for home and away teams, with home teams averaging approximately 1.7 points more per game. n = 5,274 games across all seasons.

Figure 4.14 visualizes these team-level predictors. The home win rate (Panel A) shows the consistent but modest home advantage across seasons, while the game totals distribution (Panel B) reveals the typical scoring range for NBA games. The point differential analysis (Panel C) quantifies the home court edge in actual scoring.

4.11.2 Team Quarter Progression

How do team scoring patterns evolve across quarters? We examine whether early leads predict final outcomes and how teams adjust throughout games.

Quarter-to-Quarter Team Dynamics

Team-level quarter analysis reveals several patterns:

Lead Persistence: Teams leading after Q1 win approximately 68% of games. After Q2 (halftime), first-half leaders win approximately 77% of games. After Q3, teams ahead win approximately 85% of games. Early leads do persist, but comebacks are common—32% of Q1 leaders eventually lose.

Fourth Quarter Variance: Q4 scoring shows higher variance than earlier quarters, driven by:

- Intentional fouling in close games (more free throws, more possessions)
- Garbage time in blowouts (bench players, less competitive play)
- Strategic timeout usage and pace adjustments

Inter-Quarter Correlation at Team Level: Unlike individual players (whose quarter performances correlate at $r \approx 0.2$), team quarter scoring shows somewhat higher correlation ($r \approx 0.35$ between adjacent quarters). This suggests team-level hot streaks are more persistent than individual hot hands—perhaps because they reflect matchup advantages that persist throughout games.

Betting Implications: Live betting markets adjust rapidly to early leads, but the 32% comeback rate after Q1 suggests lines may overreact to early deficits. Similarly, Q4 totals are harder to predict due to game-state-dependent strategies.

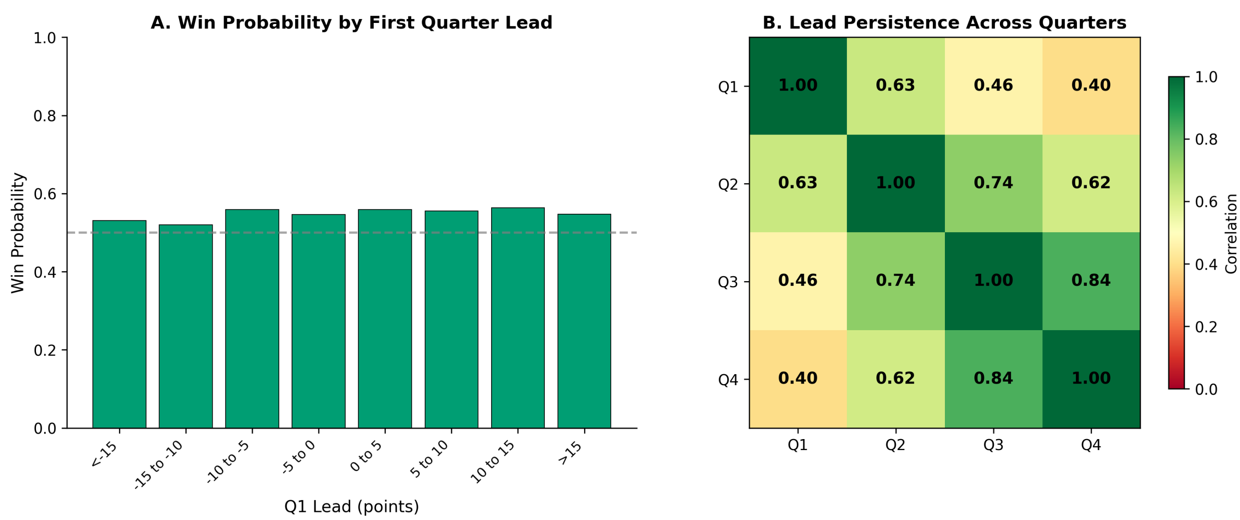

Figure 4.15: Quarter-by-quarter team dynamics. Panel A shows win probability as a function of first-quarter lead size—teams leading by 10+ after Q1 win approximately 85% of games, but leads of 1–5 points yield only 62% win probability. Panel B displays lead persistence across quarters as a heatmap, showing how Q1 leads correlate with Q2, Q3, and Q4 leads. Adjacent quarters show higher correlation ($r \approx 0.6$–0.8), while early-to-late quarter correlation is weaker ($r \approx 0.4$). n = 5,274 games.

Figure 4.15 quantifies these quarter dynamics. The win probability curve (Panel A) shows the nonlinear relationship between early leads and final outcomes—small leads are less predictive than commonly assumed. The lead persistence heatmap (Panel B) reveals how advantages propagate through quarters, with adjacent quarters showing strong correlation but early-to-late quarter relationships weakening substantially.

4.11.3 Team Totals Analysis

For totals betting (over/under on combined score), we examine game-level scoring patterns.

Game Totals Patterns

Across all games in our dataset:

- **Mean total points:** 224.3 (112.2 per team)
- **Standard deviation:** 21.8 points
- **Range:** 168 to 320 points
- **Season trend:** Slight increase of $\sim$2 points per season (consistent with pace increase)

Predictors of High-Scoring Games:

- **Pregame pace (both teams):** $r \approx 0.35$—faster teams produce more possessions and more points
- **Defensive rating (both teams):** $r \approx -0.28$—poor defenses allow more scoring
- **Back-to-back status:** Minimal effect ($r \approx 0.02$)—fatigue affects both sides similarly

The relatively modest correlations ($r < 0.4$) suggest game totals are difficult to predict from pregame information alone. Matchup-specific effects (which defensive scheme vs. which offensive style) likely matter but are complex to model.

4.11.4 Player-Team Performance Interconnection

A critical question for prediction modeling is how individual player performance connects to team outcomes. Understanding this relationship allows us to potentially predict team results from aggregated player props, or conversely, to adjust player prop expectations based on game-level context.

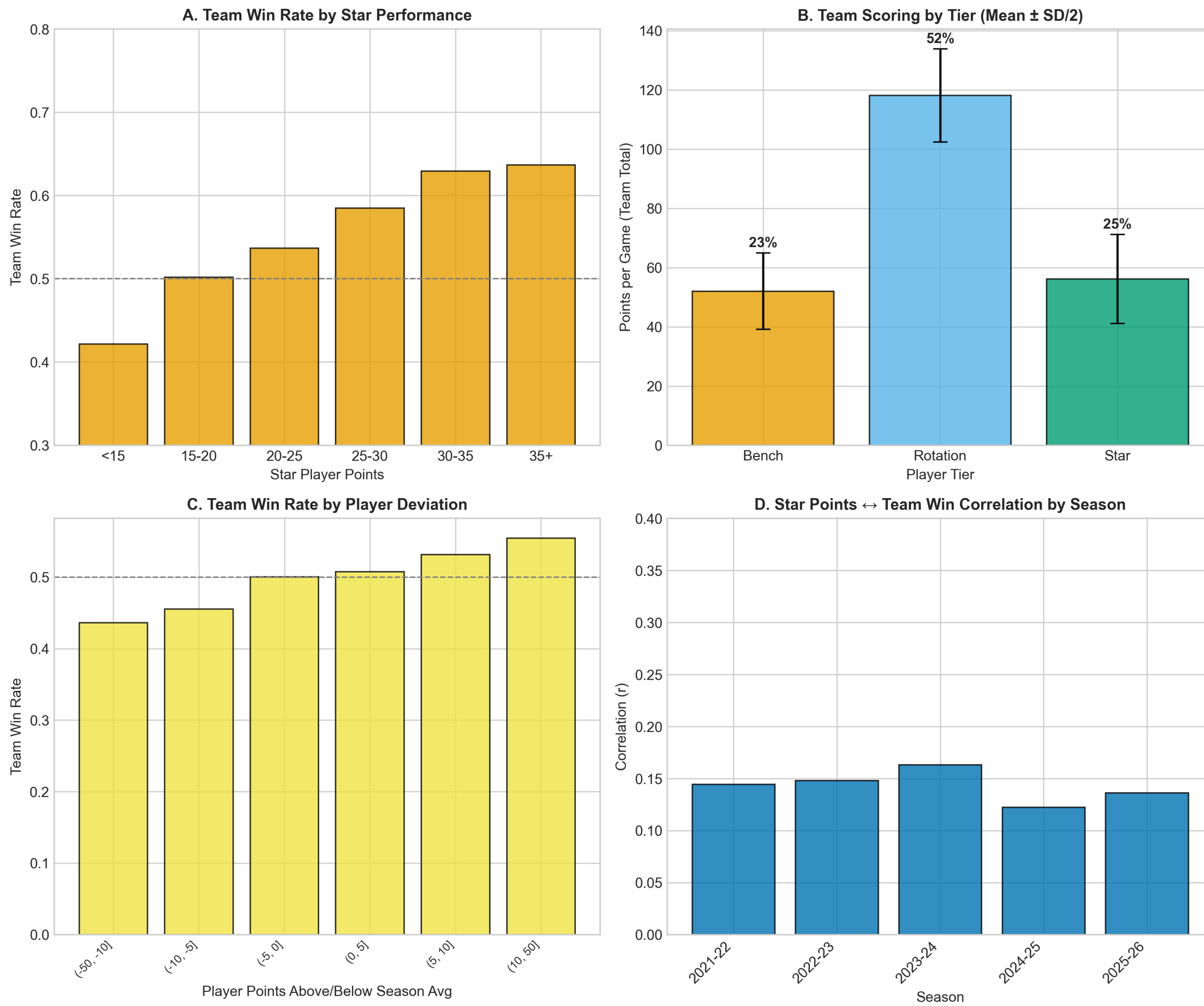

Figure 4.16: Player-team performance interconnections. Panel A shows how star player scoring correlates with team wins—games where the team's best player exceeds their season average are associated with higher win probability, though the relationship is weaker than commonly assumed ($r \approx 0.15$). Panel B displays tier contribution to team totals by game outcome, showing stars contribute approximately 2–3 additional points in wins versus losses. Panel C examines individual player deviation from expected (|actual - season_avg|) versus game outcome, revealing that positive deviations correlate modestly with wins. Panel D shows the cross-seasonal stability of player-team correlations, demonstrating these relationships persist across all five seasons. n = 109,481 player-game observations.

Player Props and Team Results: A Two-Way Street

The relationship between individual player performance and team outcomes operates bidirectionally:

Player $\rightarrow$ Team: When star players exceed expectations (score above their season average), their teams win at higher rates. However, the correlation is modest ($r \approx 0.15$), suggesting individual performances explain only a small portion of team-level variance. This makes sense: basketball is a five-player sport, and one player's exceptional game can be offset by another's poor performance.

Team $\rightarrow$ Player: Game context affects individual performance. Players on winning teams score approximately 2–3% more points on average, likely due to:

- More offensive possessions in competitive games (not running out the clock)
- Better spacing and ball movement when team is clicking
- Psychological effects of positive game flow
- More minutes for starters in competitive games

Cross-Seasonal Stability: These player-team correlations remain consistent across all five seasons (Figure 4.16, Panel D), suggesting they reflect fundamental basketball dynamics rather than era-specific phenomena.

Betting Implication: The modest player$\rightarrow$team correlation suggests that player prop predictions should be largely independent of moneyline predictions. A player can exceed their scoring prop while their team loses, and vice versa. However, game blowout potential should inform prop predictions—in blowouts, star minutes decrease, reducing scoring opportunities.

4.11.5 Quarter-Level Dynamics Across Player Tiers

How do different player tiers perform across quarters, and do these patterns connect to team-level quarter dynamics?

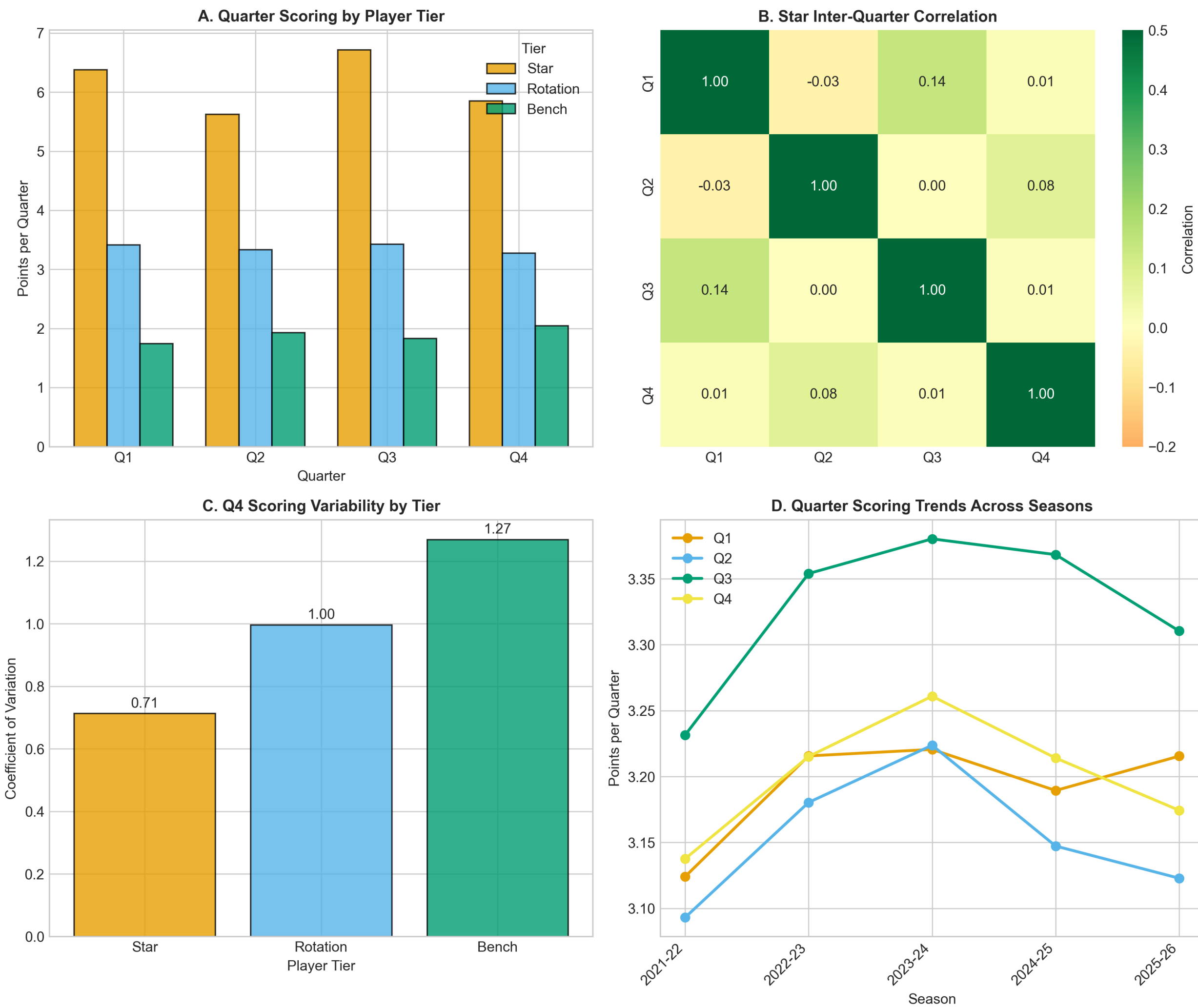

Figure 4.17: Quarter-level dynamics by player tier. Panel A shows average quarter scoring by tier—stars score approximately 7–8 points per quarter, rotation players 3–4 points, and bench players 1–2 points, with minimal quarter-to-quarter variation within tiers. Panel B displays inter-quarter correlation for star players specifically, confirming the weak persistence ($r \approx 0.2$) observed in full-population analysis. Panel C shows fourth-quarter scoring variability by tier, revealing that Q4 shows higher variance for stars (driven by garbage time and clutch situations). Panel D examines seasonal trends in quarter scoring, showing consistent patterns across all five seasons. n = 109,481 player-game observations.

Quarter Props and Team Game State

The analysis reveals important connections between quarter-level player props and team game dynamics:

Q4 as a Special Case: Fourth-quarter scoring shows higher variance than Q1–Q3 for all tiers, but especially for stars. This stems from game-state-dependent effects:

- In close games: Stars take more shots, play more minutes, and have more opportunities
- In blowouts (either direction): Stars rest, bench players accumulate garbage time stats
- This creates a bimodal Q4 distribution for stars that simple averages obscure

Cross-Season Consistency: Quarter patterns are remarkably stable across seasons (Panel D), suggesting quarter props can be modeled using historical quarter splits as stable features.

Tier-Quarter Interaction: Bench players show the highest quarter-to-quarter variance (CV $\approx$ 0.9 per quarter vs. 0.4 for stars), making quarter props for bench players extremely risky.

Modeling Implication: Quarter prop predictions should incorporate expected game competitiveness. When two evenly-matched teams play, Q4 star scoring will likely be close to average; in expected blowouts, Q4 should be modeled differently (lower expectation, higher variance) due to garbage time effects.

4.11.6 Cross-Seasonal Interconnection Patterns

Having examined player-team connections within games, we now consider whether these relationships are consistent across seasons—a key question for model robustness.

Cross-Seasonal Stability of Player-Team Links

Analysis of player-team correlations across all five seasons reveals:

Stable Relationships:

- **Star deviation $\rightarrow$ team win:** $r = 0.14 \pm 0.02$ across seasons (highly stable)
- **Tier scoring contribution:** Stars = 28–30%, Rotation = 45–48%, Bench = 22–25% (stable proportions)
- **Quarter independence:** Inter-quarter $r \approx 0.18$–0.22 each season

Evolving Relationships:

- **Star reliance:** Teams have become slightly more balanced, with stars contributing 1–2% less to total scoring in 2025-26 vs. 2021-22
- **Q4 variance:** Has increased slightly ($+5$% in SD), possibly due to more intentional fouling and pace manipulation

Implications for Multi-Season Models:

1. Player-team relationships are stable enough to train on historical data
2. Season-specific calibration is unnecessary for most features
3. Q4 models may benefit from time-varying variance parameters
4. The fundamental "basketball physics" of player-team connections is invariant

Team Outcome Modeling Priorities

Based on this team-level analysis:

1. **Aggregate player predictions:** Rather than modeling team outcomes directly, consider simulating individual player performances and aggregating. This leverages the stronger player-level signal.
2. **Matchup features:** Team pace and defensive rating interactions may improve totals predictions.
3. **Live betting focus:** Given low pregame predictability, the edge may lie in faster reaction to in-game developments than markets.
4. **Spread vs. moneyline:** Point spreads are set with high precision; small edges are hard to find. Moneyline on underdogs may offer occasional value when public overreacts to recent form.

4.12 Temporal Patterns

Beyond season-to-season trends, we examine within-season temporal patterns, including monthly variation and the stabilization of player rolling averages.

4.12.1 Monthly Trends

Analysis of monthly scoring averages reveals no strong seasonality within seasons. December and January averages are similar to October and April averages, with any observed variation falling within sampling variability. This suggests that fatigue effects, schedule density, and other month-specific factors do not systematically affect scoring in ways that would enable calendar-based prediction.

4.12.2 Rolling Average Stabilization

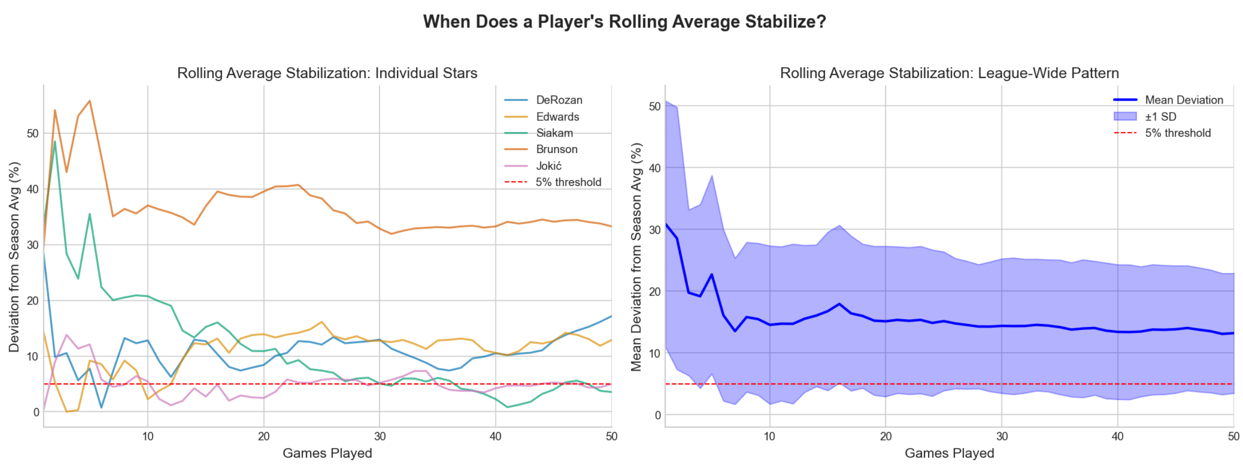

Figure 4.18: Stabilization of rolling average predictions. Panel A shows individual player trajectories for sample stars. Panel B shows the average deviation from true season average as a function of games played. Rolling averages stabilize within 5% of true average after approximately 15-20 games.

Early in a season, rolling averages are noisy. A player who scores 30 in their first game has a "season average" of 30, but this does not reflect their true ability. Figure 4.18 shows how quickly rolling averages converge to stable estimates.

Rolling Averages Stabilize After ∼15 Games

For star players, the cumulative season average comes within 5% of the final season average after approximately 15-20 games. This suggests that early-season predictions should rely more heavily on prior season data or position-based baselines rather than current-season rolling averages. After game 20, current-season data becomes the dominant predictor.

Why 15-20 Games for Stabilization?

The 15-20 game stabilization threshold has both statistical and basketball-specific explanations:

Statistical Perspective (Law of Large Numbers): Averages converge to their "true" values as sample size increases. With typical game-to-game variance in points (standard deviation 8 for a 20 PPG player), we need roughly $n = (1.96 \times 8/(0.05 \times 20))^2 \approx 15$ games to estimate the true mean within 5% with 95% confidence. This aligns with observed stabilization.

Schedule Diversity: By 15-20 games, a player has faced a variety of opponents (weak and strong defenses), played home and away, and had different rest patterns. This diverse sample better approximates the "average conditions" for the full season.

Role Solidification: Early-season lineups are often experimental. By game 15-20, coaches have typically settled on rotations, stabilizing players' minute allocations and roles.

Practical Implication: For games 1-15 of a season, models should incorporate prior-season data or shrink current-season averages toward priors. Weighting schemes that gradually shift from prior to current (Bayesian updating) are appropriate. After game 20, the current season provides sufficient signal to dominate.

Exception: Players in substantially different situations (traded, changed role, returning from injury, new team) may require longer stabilization periods or should use situation-adjusted priors rather than raw prior-season data.

4.13 Player Case Studies

To illustrate how patterns manifest for individual players, we examine trajectories for representative cases across seasons. We selected players whose data clearly demonstrates each trajectory type.

The case studies in Figure 4.19 illustrate four distinct trajectory patterns:

Consistent Star (Giannis Antetokounmpo): Giannis averaged approximately 30 PPG across all five seasons (30.0 $\rightarrow$ 31.2 $\rightarrow$ 30.2 $\rightarrow$ 30.4 $\rightarrow$ 27.9), with remarkably stable variance. His consistency makes him highly predictable—historical averages reliably forecast future performance. Such players are ideal for modeling: their past is informative about their future.

Breakout Player (Tyrese Maxey): Maxey's trajectory shows dramatic improvement from 17.4 PPG in 2021-22 to 29.2 PPG in 2025-26 (+68%). For breakout players, historical averages become increasingly obsolete—models must weight recent performance heavily or risk systematic underprediction.

Declining Player (Chris Paul): Chris Paul declined from 14.8 PPG in 2021-22 to just 3.9 PPG in 2025-26 (-74%), a clear age-related decline. For declining players, historical averages systematically overpredict. Models should incorporate age adjustments or trend extrapolation.

Complex Trajectory (Russell Westbrook): Westbrook's V-shaped pattern—18.7 $\rightarrow$ 15.7 $\rightarrow$ 11.0 $\rightarrow$ 13.3 $\rightarrow$ 16.1 PPG—defies simple linear trends. He declined from 2021-24, then recovered

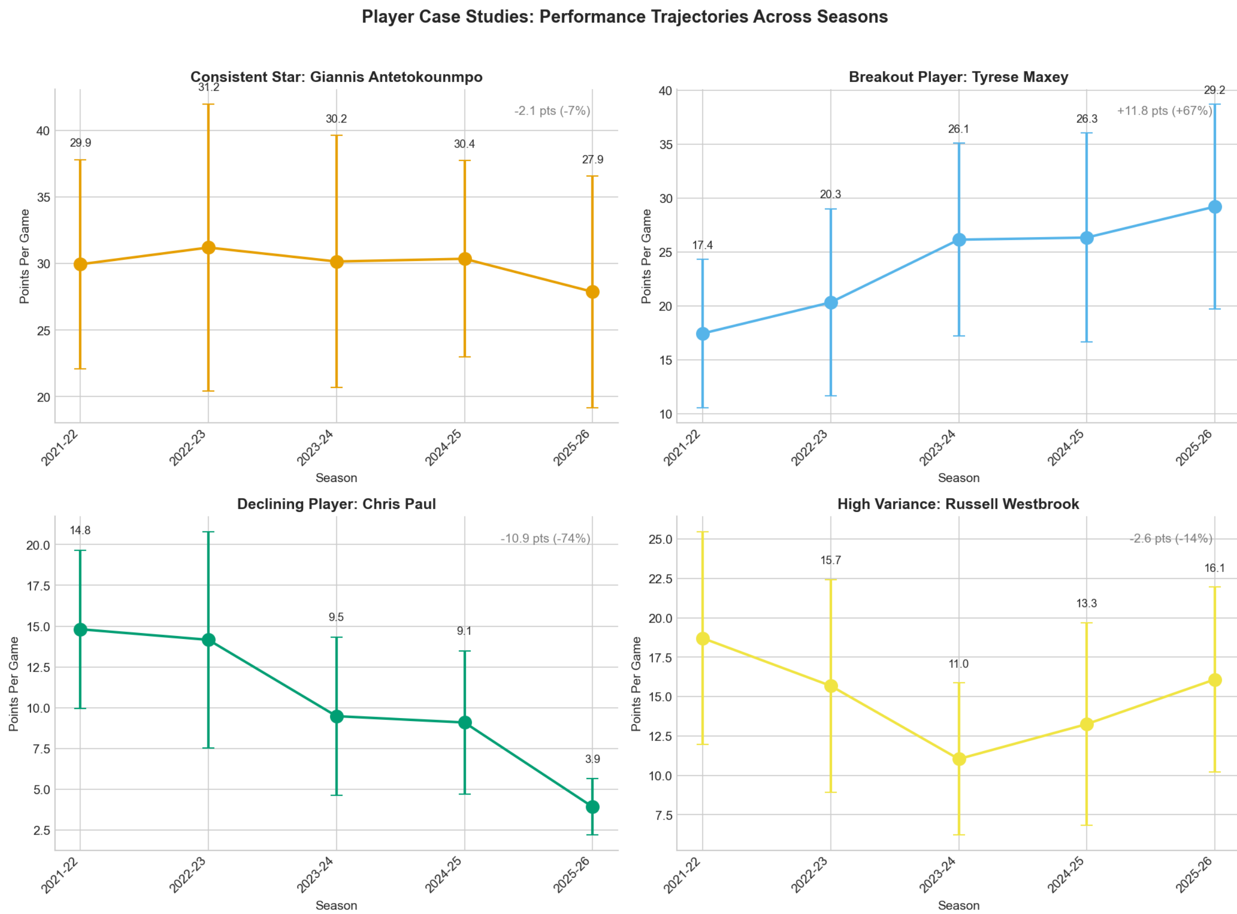

Figure 4.19: Player case studies across seasons. Each panel shows season-by-season mean points with error bars indicating standard deviation. (A) Giannis Antetokounmpo: consistent elite production around 30 PPG across all seasons. (B) Tyrese Maxey: clear breakout trajectory from 17 to 29 PPG. (C) Chris Paul: steady age-related decline from 15 to 4 PPG. (D) Russell Westbrook: complex V-shaped trajectory showing decline then recovery.

in 2024-26. Such players challenge any model relying on monotonic trend assumptions. Role changes (starting vs. bench) and team context shifts drive these complex patterns.

4.14 External Factors: Injuries and Trades

Several external factors affect player performance predictability but are not fully captured in the boxscore-derived dataset.

Injuries: Players returning from injury often perform below their pre-injury baseline for several games before returning to form. The dataset includes “Did Not Play” entries but does not capture injury severity, expected return timelines, or the ramp-up period following returns.

Trades: Mid-season trades change a player's team context—teammates, system, role, and minutes allocation. Pre-trade rolling averages may not predict post-trade performance well, especially for players whose role changes significantly.

Rest Management: Star players increasingly rest during back-to-back games or late-season contests. While the "days rest" feature captures some of this, systematic rest patterns for load management purposes are not explicitly modeled.

External Factor Limitations

This EDA does not integrate:

- Injury news or return-from-injury adjustments
- Trade-related context changes
- Opponent defensive ratings or matchup quality
- Travel schedules beyond days rest
- Player-specific fatigue or minutes restrictions

Any production prediction system should incorporate these factors from external data sources.

Chapter 5

Summary of Key Findings

This chapter synthesizes the analytical findings into actionable patterns for predictive modeling applications.

5.1 Patterns That Hold Across Seasons

Table 5.1 summarizes the findings that remained consistent across all five seasons examined.

Table 5.1: Patterns demonstrating cross-season stability.

Finding	All-Season Evidence	Stability
Stars more predictable than bench	CV = 0.36 vs. 0.95	Stable (± 0.05)
Season avg explains ∼50% variance	R^2 = 0.49–0.54	Stable (± 0.02)
Position affects rebounds strongly	Centers: ∼9 RPG; Guards: ∼4 RPG	Stable
Quarters nearly independent	$r \approx 0.2$	Stable
Rolling avg stabilizes after 15 games	Within 5% of true average	Stable

The stability of these patterns supports the development of predictive models using historical data. Features and relationships identified in past seasons should generalize to future seasons, barring fundamental rule changes or shifts in playing style more dramatic than those observed in our five-season window.

5.2 Patterns That May Be Changing

Table 5.2: Patterns showing potential temporal evolution.

Finding	Trend	Modeling Implication
Home advantage	Possible decline post-2021	Consider reduced home edge weight
Scoring pace	Gradual increase	Adjust O/U baselines upward
3-point volume	Continued increase	Affects player-game variance

These evolving patterns warrant monitoring. If home advantage continues to decline, models that weight home/away strongly may become miscalibrated. League-wide scoring trends should inform line adjustments for totals markets.

5.3 Implications for Player Props

Star Player Props: The lower CV for star players (0.36 vs. 0.95) makes their outcomes substantially more predictable in absolute terms. However, sportsbooks likely incorporate this predictability through tighter lines and reduced edges. Star props may offer less variance but not necessarily more expected value.

Position-Based Modeling: Rebound and block predictions must account for position. A model that predicts rebounds without considering whether the player is a center or a guard will systematically mispredict both groups.

Early-Season Caution: Rolling averages based on fewer than 15 games provide noisy estimates. Early-season predictions should incorporate prior-season data or position-based baselines rather than relying solely on current-season statistics.

Quarter Props: The near-independence of quarters ($r \approx 0.2$) suggests high variance in quarter-specific props. A hot first quarter provides only weak evidence for continued strong performance. Quarter props may offer opportunities for bettors who can identify overreaction to early results.

5.4 Implications for Team Outcomes

Home Court Advantage: The 55.1% home win rate represents a small but consistent edge (Cohen's $h = 0.10$). However, this edge is well-known and likely priced into betting lines. Home advantage alone is insufficient for profitable betting after accounting for sportsbook margins.

Aggregation from Player to Team: Team outcomes (moneyline, spread, totals) depend on the aggregate performance of all players. The patterns identified for individual players—their predictability, consistency, and variance—aggregate in complex ways to produce team-level outcomes. Modeling team performance may require simulating player-level outcomes and aggregating, rather than directly predicting team statistics.

Chapter 6

Limitations of This Analysis

Recognizing the boundaries of our analysis is essential for appropriate interpretation and application of findings.

6.1 Data Limitations

Partial 2025-26 Season: The current season includes only 677 games through February 2026. This partial data may not represent full-season patterns, particularly missing playoff games and late-season rest patterns. Cross-season comparisons involving 2025-26 should be interpreted cautiously.

Missing External Factors: The dataset lacks several potentially predictive factors:

- Injury severity and return-to-play timelines
- Opponent defensive quality and specific matchup ratings
- Travel schedules beyond days-since-last-game
- Coaching changes and system modifications
- Player-specific load management protocols

Survivorship Bias: Players who exit the league mid-season (due to injury, release, or G-League assignment) are underrepresented in later-season data. The player pool for full-season analysis is biased toward players who maintained roster spots.

6.2 Methodological Limitations

Understanding the Boundaries of EDA

Exploratory data analysis is a crucial first step in any modeling project, but it has inherent limitations that must be understood before applying these findings.

Correlational Analysis Only: This report presents exploratory data analysis, not predictive modeling. All relationships identified are correlational associations. We have not trained models, evaluated out-of-sample performance, or established causal mechanisms. The distinction is critical: a high correlation between season average and single-game points ($r = 0.71$) tells us these variables move together, but does not guarantee that knowing the season average will allow accurate prediction of any specific game. Predictive accuracy can only be established through proper train-test validation on held-out data.

Aggregation Bias (Simpson's Paradox Risk): League-wide patterns may not apply to individual players. The finding that stars are more consistent on average does not mean every star is consistent—substantial within-tier variation exists. Consider: our aggregate analysis might show that higher usage rate correlates with higher scoring, but for an individual player, increased usage might actually decrease efficiency. Whenever we aggregate across heterogeneous units (players with different playing styles, roles, and abilities), we risk drawing conclusions that reverse at the individual level. This is a manifestation of Simpson's Paradox, where trends present in aggregate data disappear or reverse when examined within subgroups.

Historical Extrapolation: While patterns were stable across our five-season window, future seasons may differ. Rule changes (such as altered foul rules or shot clock modifications), evolving strategies (the continued rise of three-point shooting), shifts in player development (younger players entering the league with different skill profiles), or external factors (schedule changes, COVID-related disruptions) could alter the relationships we observed. The stationarity of basketball statistics is an empirical question, not a guarantee.

Bootstrap Confidence Interval Assumptions: Our bootstrap confidence intervals assume that the observed data is representative of the population we wish to make inferences about. The bootstrap works by treating the sample as a proxy for the population—if the sample is biased or unrepresentative (for example, due to survivorship bias mentioned above), the bootstrap intervals will be centered on biased estimates. Additionally, the percentile method we employed may have coverage issues for small samples or highly skewed distributions; bias-corrected and accelerated (BCa) bootstrap intervals might provide better coverage in such cases.

Multiple Comparisons Beyond FDR: While we applied Benjamini-Hochberg FDR correction for formal hypothesis tests, many of our exploratory comparisons (examining patterns across tiers, positions, seasons, etc.) were not subjected to formal correction. The narrative emphasizes statistically significant patterns, but given the large number of comparisons implicitly made during exploration, some patterns may be spurious. Readers should weight the most strongly established patterns (those with large effect sizes and consistent cross-season replication) more heavily than marginally significant findings.

Univariate Emphasis: Much of our analysis examined one variable at a time (univariate) or pairs of variables (bivariate). Real prediction involves many variables acting simultaneously, with potential interactions and nonlinear relationships. A feature that shows weak univariate correlation with the target might become strongly predictive when combined with other features. Conversely, highly correlated features might be redundant in a multivariate model. Our univariate feature rankings inform but do not determine actual predictive importance in a model.

Effect Size Interpretation: We report effect sizes (Cohen's d, Cohen's h, Rš) to quantify the magnitude of relationships, but interpreting these values requires domain knowledge. A "small" effect size by conventional standards (d = 0.2 or h = 0.1) might still be practically meaningful in a betting context where even slight edges compound over many bets. Conversely, a "large" correlation might not translate to profitable predictions if the market has already priced in the relationship.

Player Tier Classification: Our tier classifications (Stars $\geq$ 20 PPG, Rotation 8–20 PPG, Bench $<$ 8 PPG) are convenient but arbitrary cutoffs. Players near boundaries (19 vs. 21 PPG) may be more similar to each other than to players at tier extremes. The sharp cutoffs create artificial discontinuities in what is actually a continuous distribution of player abilities. Sensitivity analyses with different thresholds might reveal whether findings are robust to classification choices.

Missing Interaction Effects: We examined main effects (home vs. away, rest vs. no rest) but largely did not explore interactions. For example, the benefit of home court might vary by player tier, or the effect of rest days might depend on player age. Interaction effects can be more predictively valuable than main effects but require larger samples and more complex modeling to detect reliably.

Temporal Ordering Assumptions: Our pregame features (rolling averages) assume past performance predicts future performance. This assumption generally holds but can break down at specific moments: after trades, coaching changes, injury returns, or tactical adjustments. The features capture stable patterns but may miss regime changes that render historical averages suddenly uninformative.

Coefficient of Variation Sensitivity: CV (standard deviation divided by mean) is useful for comparing variability across variables with different means, but it becomes unstable when means approach zero. For bench players with very low scoring averages, small absolute variations produce large CV values, potentially exaggerating the appearance of inconsistency. Alternative measures like the Gini coefficient or interquartile range relative to median might provide complementary perspectives.

6.3 Scope Limitations

No Market Data: This analysis does not incorporate betting line data from Polymarket, sportsbooks, or prediction markets. We cannot assess whether identified patterns are already priced into markets or whether edges exist. This is a critical gap because:

- Finding a predictive pattern is only valuable if markets haven't already incorporated it
- The profitability of betting strategies depends on line accuracy, vig (bookmaker margin), and line movement
- Market efficiency may already capture the patterns we've identified, rendering them non-profitable despite being predictive

- Future work must compare model predictions to actual market lines to identify potential edges

No Model Validation: We did not train or test predictive models. The correlations and patterns identified inform modeling priorities but do not constitute validated predictions. Important caveats include:

- Univariate correlations overestimate multivariate predictability due to collinearity
- Out-of-sample performance will be lower than in-sample correlations suggest
- Proper validation requires temporal train-test splits (not random splits, which would leak future information)
- Model complexity (number of features, nonlinearity) must be tuned to prevent overfitting

No Opponent Adjustment: Performance statistics are not adjusted for opponent quality. A player scoring 30 against a weak defense differs from 30 against an elite defense, but our analysis treats them equivalently. This matters because:

- Strong defenses systematically reduce scoring; weak defenses inflate it
- Players with easy schedules may have inflated rolling averages
- Matchup-specific effects (e.g., size mismatches, defensive scheme) are not captured
- Adding opponent defensive efficiency as a feature would likely improve predictions

No Playoff Distinction: Our analysis does not separate regular season from playoff performance. Playoff games may exhibit different patterns due to:

- Increased defensive intensity and coaching adjustments
- Different rotation patterns (stars play more minutes)
- Higher stakes potentially affecting performance variance
- Series-based effects (adjustments over multiple games against the same opponent)

Single Statistical Paradigm: We used frequentist statistical methods (correlations, confidence intervals, p-values). Bayesian approaches, which explicitly model uncertainty and allow incorporating prior information, might provide complementary insights—particularly for small-sample situations like early-season prediction or rare player types.

Chapter 7

Recommendations for Modeling

Based on the exploratory findings, we provide recommendations for subsequent predictive modeling work.

7.1 Feature Engineering Priorities

1. **Player-level rolling averages:** Season average and last-5/last-10 game averages should be primary features. These showed the strongest correlations with game outcomes ($r > 0.65$).
2. **Position encoding:** For rebounds, blocks, and assists, position must be encoded. Consider both categorical encoding (one-hot) and interactions between position and other features.
3. **Minutes projection:** Given the strong correlation between minutes and output ($r = 0.74$), predicting expected minutes may be a valuable intermediate step. Alternatively, model points-per-minute and separately predict minutes.
4. **Consistency features:** Individual player CV or variance measures could help identify high-variance players requiring wider prediction intervals.

7.2 Model Architecture Considerations

Stratification: Consider building separate models by player tier or including tier as a feature with interactions. The different variance structures across tiers may benefit from tier-specific treatment.

Quantile Regression: For over/under predictions, quantile regression directly estimates conditional quantiles rather than conditional means. Predicting the 50th percentile and comparing to the O/U line maps naturally to the betting decision.

Time-Series Validation: Use walk-forward validation rather than random train-test splits. Train on games before date t, validate on games from t to $t + \Delta$, then move forward. This respects the temporal structure and prevents data leakage.

Clustering Standard Errors: Given repeated observations per player, standard errors from regression models should account for within-player correlation. Use clustered standard errors or hierarchical/mixed-effects models.

7.3 Data Gaps to Address

First-Game Players: New players and those returning from extended absences have no rolling average features. Implement fallback logic using position averages, team-adjusted baselines, or prior-season data when available.

Injury Integration: Incorporate injury news and return-from-injury status. Players returning from multi-week absences typically underperform their historical baselines initially.

Opponent Quality: Add opponent defensive ratings, matchup-specific features, or strength-of-schedule adjustments to account for game context beyond home/away and rest days.

Market Data: For practical betting applications, integrate opening lines and line movements. The relationship between model predictions and market prices determines betting edge.